

Countercyclical Bank Equity Issuance

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Over the period 1980–2012, large U.S. commercial banks raise and retain less equity during credit expansions, which amplifies their leverage. The decrease in equity issuance is large relative to subsequent banking losses. I consider a variety of explanations for why banks resist raising equity and find evidence consistent with the diminishment of creditor market discipline due to government guarantees. I test this explanation by analyzing the removal of government guarantees to German Landesbank creditors and find that creditor market discipline and equity issuance increase. These findings help explain why banks resist raising equity, making financial distress more likely. (*JEL* G21, G28, G32, G35)

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A central issue in banking finance concerns the optimal capital levels of banks and whether bank equity is expensive from both social and private points of view. It is often argued that higher capital requirements may reduce banks' risk-taking incentives and help banks better withstand adverse shocks (Admati et al. 2013). In particular, regulators have stressed higher capital requirements during *credit expansions* due to the increased risk of financial instability: credit expansions predict increased credit losses and risk of banking crises (Schularick and Taylor 2012), negative returns for bank stocks (Baron and Xiong 2017), and economic downturns (Mian, Sufi, and Verner 2017). As a result, policy makers have argued for policies including countercyclical capital buffers and dynamic provisioning that would implement higher capital requirements during credit expansions.

This paper finds that for large U.S. commercial banks over the period 1980–2012, banks raise and retain *less* common equity during credit expansions,

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even though more equity might help banks to better absorb subsequent shocks. The reduced equity issuance and retained earnings during credit expansions is large in magnitude relative to subsequent bank losses. For example, in the most recent credit expansion (2004–2007), the decreased net equity issuance (relative to trend) was more than \$200 billion and dividend payouts were in excess of \$450 billion for the twenty largest U.S. banks, compared to roughly \$500 billion in subsequent losses for these banks during the 2007–2008 financial crisis. Thus, in the counterfactual in which banks substantially reduce payouts and do not reduce new issuance during the boom, banks would have had more than sufficient buffer to withstand subsequent credit losses. Banks instead wait until after the downturn to raise equity, when it is potentially costlier and too late to mitigate a crisis. This is a robust pattern I label “countercyclical bank equity issuance.”

If credit expansions predict credit losses and bank distress, why do large banks postpone raising equity or retaining income during credit expansions, often to be forced by regulators or markets to recapitalize during the downturn, when conditions for issuing equity are much less favorable? Issuing equity during a downturn can be costly due to increased debt overhang, “stigma” (equity issuance as a negative signal), and depressed share prices. In an otherwise frictionless setting, these considerations would presumably lead forward-looking bankers to raise equity in advance, during the expansion. In fact, most models imply that bank equity should vary *procyclically* across credit cycles: for example, if banks hold minimal required regulatory capital ratios, then equity will increase proportionately with assets in expansions and will thus be procyclical. In addition, the procyclicality of bank equity would make sense, both because bank income is high during credit expansions, allowing banks to use retained earnings as an efficient way to increase equity, and because bank stock prices are high in good times (Baron and Xiong 2017), making equity issuance attractive from a market timing point of view.

Why does the logic of procyclical equity issuance not hold for large U.S. banks? This paper argues that government guarantees to bank creditors are a quantitatively important friction that leads banks to resist issuing equity during credit expansions.¹

This paper first presents a simple three-period model of bank size and capital structure to help explain how government guarantees can interact with agency frictions to produce different patterns of equity issuance and payouts across credit cycles. In the model, bankers (working in the interests of shareholders) try to take advantage of shareholder limited liability by

¹ Of course, the observed patterns of equity issuance may be driven by other factors too. This paper focuses on creditor guarantees as one important friction, as they are the subject of considerable policy debate and are shown here to be important in magnitude. Other factors, which may potentially help explain patterns of equity issuance and are discussed in Section 5, include bank shareholder governance; bank manager overoptimism, or short-termism (Malmendier and Tate 2005); value-at-risk, or return-on-equity targeting (Adrian and Shin 2013); market timing (Baker and Wurgler 2002); and more.

postponing new equity issuance—or even increasing current equity payouts, if the risk-shifting incentives are great enough—during a bank asset expansion. However, *without creditor guarantees*, bankers are prevented from doing so, because creditors, wanting to reduce the probability of bankruptcy, threaten to raise debt funding costs in the event of equity payouts. This threat of increased funding costs, which I call “creditor market discipline,” leads bankers in equilibrium to raise new equity during the credit expansion, hence leading to procyclical equity issuance. However, in the presence of government guarantees to creditors, creditor market discipline is blunted, allowing bankers to risk-shift by postponing equity issuance and increasing payouts. However, postponing equity issuance increases the chances the bank will later need to engage in a costly and inefficient recapitalization if the bank falls below minimum capital requirement—these dynamics produce the countercyclical pattern. With government guarantees, large banks resist raising equity because equity issuance dilutes the option value of equity without simultaneously lowering debt funding costs, making equity issuance costly for shareholders.

The empirical evidence linking creditor guarantees and equity issuance proceeds as follows. I first show, in a bank-level panel regression, that net equity issuance by large U.S. banks over the period 1980–2012 is countercyclical. Sorting banks by size, I show a robust size factor: the largest banks are the most strongly countercyclical, while smaller banks display a less countercyclical or acyclical pattern—and these patterns are sizeable in economic magnitudes, relative to subsequent banking losses. I also provide evidence that smaller banks raise equity often and in relatively large amounts—contrary to assumptions that banks are often resistant to raising new equity and do so rarely (e.g., Admati et al. 2013; Adrian and Shin 2013). This last finding further reinforces the notion that, whatever frictions prevent banks from raising new equity during credit expansions, these frictions are most pronounced for the largest banks.

I then explore various channels that might explain the size factor, including the potential role of government guarantees to creditors. I use Fitch support ratings, a bank-specific annual measure of expected government support to banks, which is commonly used as a proxy for perceptions of creditor guarantees and government support (e.g., Gropp, Hakenes, and Schnabel 2013). Using this measure, I show that bank equity issuance is most countercyclical when expected government guarantees are strongest. These results hold even after controlling for bank-level characteristics (e.g., net income, market-to-book, past stock returns, and a measure of undercapitalization), along with other characteristics that may affect equity issuance and payouts, such as measures of bank risk-taking, access to wholesale funding (Damar, Meh, and Terajima 2013), and geographical asset diversification (Demsetz and Strahan 1997).

While the above evidence implicates the potential importance of government guarantees, one may be concerned about potential endogeneity. I thus turn to a quasi-experimental setup, in which I exploit the surprise removal of

government guarantees to German Landesbank creditors on July 17, 2001. Up until 2001, the Landesbanken enjoyed an explicit guarantee on all bank liabilities (“Gewährträgerhaftung”), which provided a competitive advantage to the bank through high credit ratings and low funding costs.² Following a European Commission investigation into illegal state aid, the European Commission issued a surprise ruling slowly phasing out creditor guarantees according to a scheme based on a bond’s issuance date and maturity: liabilities issued between July 19, 2001, and July 18, 2005, and maturing no later than December 31, 2015, were still covered by the guarantor’s liability, whereas those issued until July 18, 2005, but maturing after December 31, 2015, were not guaranteed; and liabilities issued before the July 18, 2001, agreement were grandfathered in and maintained their guarantees, regardless of their maturities. I exploit the discontinuities inherent in these rules, along the fact that the removal of creditor guarantees had differential effect across institutions, to identify the consequences of removing creditor guarantees.

Using this setup, I first test the prediction that government guarantees dull creditor market discipline in response to equity issuance and payout announcements. I show that, without creditor guarantees, subordinated debt yields are sensitive to equity issuance and payout announcements. Then, I demonstrate the causal effect of the removal of government guarantees for the German Landesbanken, which leads creditor markets to be more sensitive to equity issuance and payouts after government guarantees are removed. Exploiting the rules from the European Commission ruling, exogenous variation comes from comparing specific bond issues at each bank that were grandfathered into the old system and thus had government guarantees, versus newer bond issues (of similar seniority, modified duration, and other characteristics) that did not, and comparing the reaction of debt yields to equity issuance and payout announcements. I find a 28-basis-point (bp) drop in marginal debt funding costs subsequent to each equity issuance (and similar changes in the opposite direction subsequent to share repurchases and dividend increases); in contrast, guaranteed bank debt is insensitive to equity issuance and payouts. The result suggests that banks may resist raising equity because, with government guarantees, equity issuance is not compensated with a decrease in debt funding costs, making equity issuance costly for shareholders.

I then analyze the response of the German Landesbanken in terms of equity issuance and retention. I find that the banks most affected by the European Commission ruling (i.e., those that lose a greater fraction of creditor guarantees) raise more equity and reduce payouts over the subsequent Landesbank credit expansion. Exogenous variation comes again from the fact that bonds issued before the ruling were grandfathered in and continued to have government

² The Landesbanken, which will be further described in Section 4.1, are wholesale and investment banks, affiliated and partially owned by German federal states, that both lend to corporate customers and act as liquidity clearinghouses for public savings banks.

guarantees, and the rate at which guaranteed debt is phased out for a given bank depends on idiosyncratic differences with the maturity structure of its debt: banks with longer maturity debt pre-2001 have their guarantees phased out more slowly because of the longer time before they have to rollover their debt. Specifically, I find that when the fraction of guaranteed debt decreases by 10 percentage points, cumulative equity issuance over the subsequent credit cycle increases by 23.36 percentage points of book equity, and retained earnings increase by 13.57 percentage points of book equity. These results suggest that creditor guarantees have a meaningful impact on the decision by bankers to issue new common equity during credit expansions.

This paper builds on Acharya et al. (2012) and Acharya, Le, and Shin (2017), who study bank payouts from 2002 to 2012 and find that payouts increased during the 2002–2008 credit expansion, and especially during the first half of 2008. For the largest banks, my empirical results are consistent with theirs.³ However, the main difference is that their results are a special case of mine—specifically for large banks, those presumably with implicit creditor guarantees, to which they restrict their analysis. They attribute their findings mainly to agency frictions and dividend complementarities between banks, which is consistent with my framework (with dividend complementarities serving as an amplifier of the agency frictions in my model). However, their model does not include a role for creditor discipline, which I show can prevent excessive dividend payouts in smaller banks.

This paper is also related to Adrian and Shin (2013), who argue that the equity of large U.S. banks is “sticky” and nearly all their change in leverage (on an absolute dollar basis) is driven by change in debt and deposits. In Online Appendix Section G, I demonstrate consistency by replicating their results with my data set but show why countercyclical equity issuance is difficult to detect in the way visualized by Adrian and Shin (2013). Additionally, they restrict their analysis to the eight largest commercial and investment banks, whereas I show that smaller banks raise equity often and in relatively large amounts.

Finally, this paper is related to the extensive literature on capital structure and capital requirements of banks. Kashyap, Rajan, and Stein (2008) provide an overview of various perspectives motivating bank capital requirements. Theories of bank capital structure have generally focused on the role of capital structure in disciplining bank managers (e.g., Calomiris and Kahn 1991; Diamond and Rajan 2000, 2001), the benefits of liquidity creation (e.g., Kashyap, Rajan, and Stein 2002; DeAngelo and Stulz 2013), asset substitution and monitoring of borrowers (e.g., Holmstrom and Tirole 1997; Mehran and Thakor 2010), and competition on the lending side (e.g., Allen, Carletti, and Marquez 2011). This paper, in contrast, highlights the

³ In Online Appendix Section G, I replicate their results with my data, finding that net equity issuance minus payouts is negative from 2002 to the first half of 2008 (during the credit expansion) and then positive in 2009 to 2010 (after bank losses are realized), consistent with the countercyclical pattern.

key role of government guarantees to creditors as a quantitatively important friction affecting bank capital structure. While this idea has been discussed in the past, this is the first paper that directly tests this hypothesis and assesses its implications.⁴ Most importantly, this paper is the first to show that the capital structure distortions due to government guarantees may be particularly severe during credit booms, when the risk to financial stability is already elevated.

1. Theory on the Role of Creditor Guarantees

1.1 Model overview

The model has three periods and two types of agents: bankers (working in the interest of shareholders) and creditors, both of which are risk neutral. At $t=0$, the initial equilibrium capital structure of the bank is determined as follows: creditors choose an interest rate r at which to lend to the bank, and bankers determine the asset size of the bank and the ratio of liabilities to assets. At $t=1$, there is an exogenous and unanticipated shock to the opportunity cost (δ) of the bank creditors (interpreted as a “credit supply shock”), which decreases debt funding costs for the bank. This shock at $t=1$ leads bankers and creditors to reoptimize their choices of assets, liabilities, and funding costs in equilibrium, which generates an asset expansion funded by a mix of new debt and equity. This allows us to characterize equity issuance during the asset expansion. At $t=2$, the random returns on the bank’s assets are realized.⁵ If the bank’s equity falls below zero, then the bank declares bankruptcy, meaning equity shares are worth zero while the remaining losses are absorbed $(1-\gamma)$ by creditors and γ by the government. If, instead, the bank’s equity does not fall below zero but falls below a fraction Φ of assets (interpreted as minimum capital requirements), the bank is forced to engage in a costly recapitalization, forcing it to increase its equity to this minimum level Φ . These two possibilities, bankruptcy or a costly required recapitalization, are known in advance to bankers and creditors at $t=0$ and $t=1$, who take them into account in their decisions.⁶

⁴ Miller (1995) shows that depositor guarantees do *not* alone break the Modigliani-Miller theorem. See the discussion in Mehran and Thakor (2010). This paper’s mechanism thus stresses the interaction between agency frictions and guaranteed debt.

⁵ In terms of mapping the timing of the model to the data, one can roughly think of $t=0$ (initial period), $t=1$ (credit boom), and $t=2$ (returns realized, recapitalization) as 2002–2003, 2004–first half of 2008, and second half of 2008–2010, respectively.

⁶ Extensive evidence suggests that credit expansions may be driven, in part, by overoptimism and that bankers may not always maximize shareholder value during such periods (Baron and Xiong 2017; Cheng, Raina, and Xiong 2014; Fahlenbrach, Prilmeier, and Stulz 2017). The model setup—featuring shareholder-maximizing bankers and a shareholder-creditor conflict—is simply a convenient mechanism to generate a credit boom in a tightly specified model. Regardless of whether the credit boom is driven by overoptimism or agency conflicts, the key force of the model that explains the cross-section of banks is “creditor market discipline.” That is because, as demonstrated empirically, the main cross-sectional characteristic that explains the cyclicality of equity issuance is *bank size*. Thus, to explain the cross-sectional pattern, one needs to invoke the *interaction* of overoptimism (which is uniformly present across the bank size distribution) and creditor market discipline (which is present for smaller banks but dulled for larger banks due to perceptions of government guarantees). This is because, even

By featuring an asset expansion in $t = 1$, the model can be used to address several key questions in this paper. To what extent do bankers raise new equity (versus debt) to fund an asset expansion driven by an exogenous credit supply shock? Do bankers raise equity in advance (at $t = 1$), or do they wait until after a negative shock is realized (at $t = 2$), at which point they are forced by regulators to raise equity? And, lastly, how do government guarantees to creditors affect these decisions?

For certain parameters, especially when government guarantees to creditors are weak or absent, the model shows that a procyclical pattern of equity issuance emerges, as creditor market discipline forces bankers to raise equity in advance during the boom ($t = 1$), requiring less mandatory equity issuance (to meet minimum capital requirements) if things go bust (at $t = 2$). Creditors prefer to raise bank equity sooner ($t = 1$) to minimize the probability of future bankruptcy. In the absence of government guarantees, they force bankers to do so in equilibrium by threatening to raise funding costs for the bank. In fact, in the model's socially optimal benchmark which maximizes total expected bank value, the bank raises substantial new equity in advance during the boom to minimize the probability of future bankruptcy.

However, for other parameter values, especially when government guarantees to creditors are strong, a countercyclical pattern of equity issuance emerges from the model. The intuition is that, with dulled pushback from creditors due to their government guarantees, bankers will exploit their *option* of postponing equity issuance, which arises due to limited liability. Freed from creditor discipline, bankers will thus exploit their option of postponing equity issuance during the boom (at $t = 1$)—potentially even depleting equity through increased payouts, if the risk-shifting incentives are strong enough—increasing the chances that they will have to raise new equity at $t = 2$ to meet minimum capital requirements. These patterns will manifest as countercyclical equity issuance.

Note that the countercyclical pattern is, in part, driven by required recapitalization during the bust to meet minimum regulatory capital requirements – this feature is built into the model by assumption. Instead, the issue addressed by the model is, if banks anticipate that the government will force costly equity issuance in bad times, why not issue equity in advance during the boom when conditions are more favorable? This model shows that the decision to raise equity in advance during the boom depends on creditor market discipline to force banks to raise equity ahead of time, which in turn depends on the intensity of government guarantees to creditors (γ).⁷

in the presence of banker overoptimism, creditors of small banks can “apply the brakes” by requiring higher funding costs, if bankers do not raise more equity. In contrast, large banks face smaller constraints from their creditors due to perceptions of government guarantees and can thus rapidly expand their balance sheets, even while simultaneously making large equity payouts.

⁷ Why is this effect specific to credit expansions? The model highlights a nonlinearity: during a credit expansion driven by an exogenous “credit supply shock,” the decrease in debt funding costs leads to an asset expansion that is

1.2 Competitive and socially optimal allocations

Online Appendix Section A offers the model's setup, timing, and solution. Here I briefly compare the competitive allocation to the socially optimal benchmark; and in the next subsection, I further illustrate some of the intuition of the model results through comparative statics and numerical simulation of the competitive allocation.

In the competitive allocation, bank asset size (A), debt liabilities (L), and equity (E) are given by Equations (A9) and (A13). These equations apply both at $t=0$ (i.e., initial bank capital structure) and at $t=1$ (i.e., after the exogenous and unexpected "credit supply shock"), which characterize equity issuance during the resultant asset expansion. After the random shock is realized to bank asset returns at the start of $t=2$, the required recapitalization (in expectation) at $t=2$ is given by Equation (A14).

One can define the socially optimal allocation as the values of r , L , and A that maximize total firm value. Total firm value is defined as the sum of the bankers', creditors', and government's expected values together. Intuitively, it is easy to see that the socially optimal allocation is equivalent to the competitive market allocation with government guarantees set to zero ($\gamma=0$). The reason is that, in the market allocation with no credit guarantees ($\gamma=0$), the bank's benefit from exploiting the option value of equity from limited liability is perfectly offset by the increased debt funding cost r charged by bank creditors. Without government guarantees ($\gamma=0$), creditors perfectly enforce market discipline: if bankers attempt to risk-shift by increasing leverage, creditors will threaten higher funding costs, which will make the bankers perfectly internalize the costs of risk-shifting that they impose on creditors and the government.

Thus, the socially optimal allocations are given by Equations (A9) and (A13) with $\gamma=0$. Comparing the competitive market allocations to the socially optimal allocation uncovers two inefficiencies resulting from government guarantees to creditors. Relative to the socially optimal benchmark, the bank is (1) too large, resulting in a negative marginal return on assets, and (2) too leveraged, resulting in an increased default probability and expected bailout costs borne by the government. In the next subsection, I quantitatively examine these distortions.

1.3 Dynamics of equity issuance during credit expansions

I quantitatively calibrate the model and characterize patterns of equity issuance during the asset expansion at $t=1$ and the potential subsequent bust at $t=2$.

initially more tilted toward debt (because of the decrease in debt funding costs), which in turn exacerbates agency frictions (because higher leverage increases the option-value of equity), which further tilts the capital structure toward debt and further increases agency frictions—and this process continues in an amplifying fashion, to the point at which equity can actually *decrease* as assets are growing, for certain parameter values. Note, however, that the observed dynamics come in part simply from the choice to model and empirically examine the credit boom happening *first* (because of the exogenous and unexpected "credit supply shock"), followed by a potential credit bust conditional on bad asset return realizations. One can alternatively run the model in "reverse": modeling an exogenous "credit supply tightening," which would lead to an asset contraction and bank deleveraging, with the bank capital structure tilting more toward equity in this case, through the same mechanisms.

For certain parameters, especially when government guarantees to creditors are weak or absent, the model shows that a procyclical pattern of equity issuance emerges, as creditor market discipline forces bankers to raise equity in advance during the asset expansion (at $t = 1$), necessitating less required equity issuance during a potential bust (at $t = 2$) to meet minimum capital requirements. However, for other parameter values, especially when government guarantees to creditors are strong, bankers postpone equity issuance, and even deplete equity, during the boom (at $t = 1$), increasing the probability of a forced recapitalization to meet minimum capital requirements, if things go bust (at $t = 2$); these forces will manifest as countercyclical equity issuance.

In the following calibration, assume the following parameters.⁸ To assess the role of government guarantees and their effect on patterns of equity issuance over the cycle, one can vary the strength of creditor guarantees γ from $\gamma = 0.2$ representing weak guarantees to $\gamma = 0.7$ representing strong guarantees (at which point the bank's leverage increases beyond that allowed by minimum capital requirements).

Figure 1 plots patterns of equity issuance from $t = 0$ to $t = 2$ for various levels of government guarantees γ from 0.2 to 0.7. It is assumed that, at $t = 1$, there is an unanticipated and exogenous increase in δ from 0.02 to 0.04, which represents a "credit supply shock," which leads to the bank asset expansion, which is funded by a mix of new debt and equity issuance. Figure 1 plots the increase (or decrease) in net equity at $t = 1$, which is calculated from Equations (A9) and (A13). Figure 1 also plots the required recapitalization (in expectation) at $t = 2$, calculated from Equation (A14).

Figure 1 thus shows the main prediction of the model. As one can see, for weak government guarantees (e.g., $\gamma = 0.2$), the pattern of equity issuance is procyclical, with greater equity issuance at $t = 1$ and relatively less (in expectation) at $t = 2$. The opposite pattern emerges for strong government guarantees (e.g., $\gamma = 0.7$), where equity issuance is highly countercyclical: large net payouts of almost 20% of initial equity at $t = 1$, followed by greater expected required net equity issuance at $t = 2$.

To better understand the model's predictions, I also plot various comparative statistics in Figure 2. Panels A and B plot the bank's asset size and debt-to-asset ratio, conditional on several values of government guarantees γ ranging from 0.2 to 0.7. Recall that the asset expansion between $t = 0$ and $t = 1$ is driven by an unanticipated and exogenous decrease in bank funding costs, δ : therefore, one can trace the parameter δ on the x -axis left-to-right, as it increases from its initial value of $\delta = 0.02$ at $t = 0$ to various values ranging from $\delta = 0.02$ to $\delta = 0.05$ at $t = 1$, corresponding to a 0- to 3-percentage-point decrease in bank

⁸ Let bank risk be given by $\sigma = 0.4$, the initial benefit of debt at $t = 0$ to be $\delta = 0.02$, the recapitalization cost $c = 20$, minimum capital requirements $\Phi = 0.06$, and the franchise value-to-asset ratio $(V/A) = 0.05$. For the bank's production function $f(A)$, normalize $A_0 = 1$ and assume $\alpha = 0.95$.

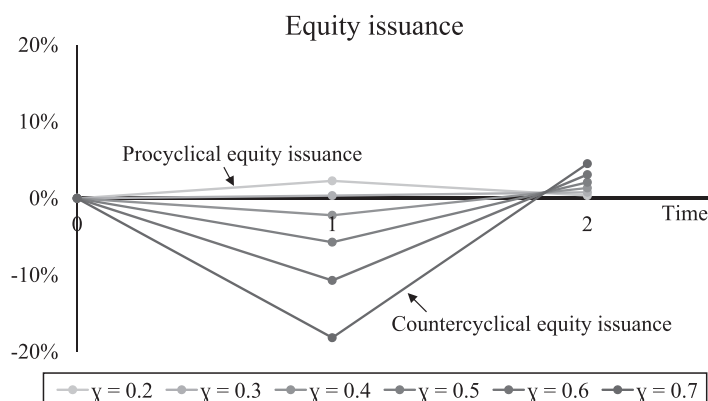


Figure 1
Model-based results: Cyclicity of bank equity issuance

Figure 1 plots patterns of equity issuance from $t=0$ to $t=2$ for various levels of creditor guarantees γ (from $\gamma=0.2$ representing weak guarantees to $\gamma=0.7$ representing strong guarantees). In the model, at $t=1$, there is an unanticipated and exogenous increase in δ from 0.02 to 0.04, which represents a “credit supply shock” and leads to a bank asset expansion that is funded by a mix of new debt and equity issuance; the figure plots the increase (or decrease) in net equity at $t=1$ calculated from Equations (A9) and (A13). The figure also plots the expected net equity issuance at $t=2$ calculated from Equation (A14). The main predictions of the model can be seen in the figure: for weak government guarantees (e.g., $\gamma=0.2$), equity issuance is procyclical, with greater equity issuance at $t=1$ and relatively less (in expectation) at $t=2$. The opposite pattern emerges for strong government guarantees (e.g., $\gamma=0.7$): equity issuance becomes highly countercyclical, with large net payouts at $t=1$ of almost -20% of initial equity, followed by greater expected required equity issuance at $t=2$.

funding costs. The effect of the asset expansion on bank asset size and its debt-to-asset ratio can therefore be gauged from tracing a given plot from left to right (i.e., from $\delta=0.02$ at $t=0$, to values $\delta \geq 0.02$ at $t=1$, with the increase in δ corresponding to the magnitude of the “credit supply shock”). Various parameter values of creditor guarantees (from $\gamma=0.2$ to $\gamma=0.7$) correspond to the variously colored lines. Figure 2 shows that higher levels of government guarantees (γ) are associated both with greater initial bank size and leverage and a greater increase in those quantities during the credit expansion as δ increases.

Similarly, panel C looks at net equity issuance (relative to initial equity when $\delta=0.02$), conditional on several values of γ . As one can see, net equity issuance is procyclical (i.e., positive as δ increases) when $\gamma=0.2$ (weak creditor guarantees) and strongly countercyclical (i.e., negative as δ increases) when $\gamma=0.7$ (strong creditor guarantees). Furthermore, the net equity issuance is nonlinear in the size of the expansion: as δ increases, the net payouts increase in an accelerating pattern. The reason, as explained in a previous footnote, is that a large credit expansion funded mainly by debt will increase the option-value of equity, further leading bankers to increase bank leverage in an amplifying manner.

To quantitatively assess the distortions arising from government guarantees and assess the welfare consequences, Figure 2 compares outcomes with low government guarantees ($\gamma=0.2$) versus those with high government guarantees

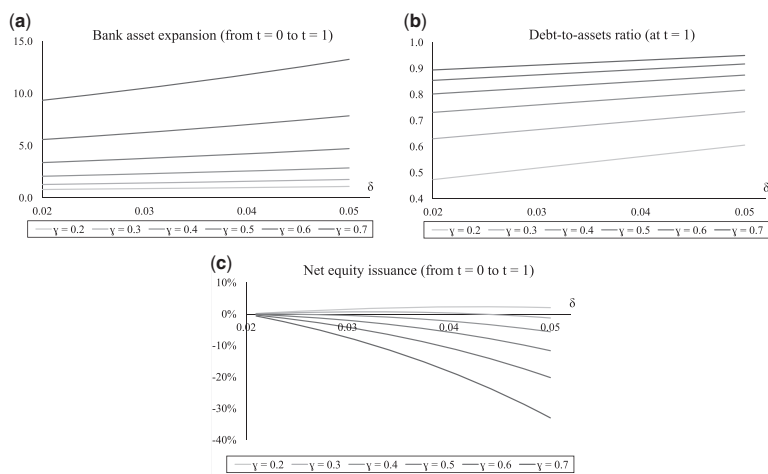


Figure 2
Model-based results: Comparative statics

Figure 2 plots the bank asset expansion (from $t=0$ to $t=1$) in panel A, the debt-to-asset ratio (at $t=1$) in panel B, and net equity issuance (from $t=0$ to $t=1$) in panel C, as functions of the parameters δ (which captures the decrease in bank funding costs due to the “credit supply shock” at $t=1$) and γ (which captures the strength of creditor guarantees). In the model, the asset expansion at $t=1$ is driven by an unanticipated and exogenous decrease in bank funding costs at $t=1$, modeled by having δ (plotted on the x-axis) increase from its initial value of $\delta=0.02$ to various values ranging from $\delta=0.02$ to $\delta=0.05$. The results of the model are plotted for various values of the parameter γ (the strength of creditor guarantees) ranging from $\gamma=0.2$ to $\gamma=0.7$ (corresponding to the variously colored lines). The plots can be interpreted as follows: the size of the bank asset expansion, change in the debt-to-asset ratios, and net equity issuance from $t=0$ to $t=1$ can be gauged from tracing a given line from left to right (given that δ is initially at $\delta=0.02$ at $t=0$ and then increases at $t=1$ to $\delta \geq 0.02$, with a larger increase in δ corresponding to a greater “credit supply shock”). Different lines correspond to different parameters values of γ (the strength of creditor guarantees): higher levels of government guarantees (γ) are associated both with higher *initial* levels (i.e., when $\delta=0.02$) of asset size (panel A) and leverage (panel B) and with greater increases in asset size and leverage during the credit expansion at $t=1$ (i.e., when $\delta \geq 0.02$). One can also see from panel C that net equity issuance is procyclical when $\gamma=0.2$ and strongly countercyclical as γ approaches 0.7. Furthermore, net equity issuance is nonlinear in the size of the expansion (i.e., as a function of δ).

($\gamma=0.7$).⁹ Panel A shows that, for the initial bank capital structure (at $t=0$), bank asset size is 9.359 with high guarantees ($\gamma=0.7$) versus 0.809 with low guarantees ($\gamma=0.2$); panel B shows similarly that the debt-to-asset ratio is 0.895 with high guarantees ($\gamma=0.7$) versus 0.474 with low guarantees ($\gamma=0.2$). To assess the magnitude of equity issuance and payouts, panel C shows that, during an asset expansion at $t=1$ (modeled by letting δ increase from 0.02 to 0.04), net equity issuance is -18.18% (“countercyclical”) with high guarantees versus $+2.28\%$ (“procyclical”) with low guarantees.

Finally, I report other calculations on the welfare consequences of government guarantees. According to these additional calculations (not

⁹ In the socially optimal benchmark with $\gamma=0$, the bank is entirely equity funded, which is perhaps a bit too extreme to serve as a realistic benchmark. Therefore, I use $\gamma=0.2$ as the “near-socially-optimal” benchmark, given that almost all banks have some sort of basic government guarantees like deposit insurance. It is interesting, however, to note that a bank with $\gamma=0$ is about 9 times smaller in assets than when $\gamma=0.7$, demonstrating how creditor guarantees can greatly increase a bank’s size, perhaps further increasing perceptions of being “too big to fail.”

reported in the figures), the total social value created (the sum of the bankers', creditors', and government's expected benefits) as a fraction of initial assets is -16.4% with high guarantees ($\gamma=0.7$) versus $+2.0\%$ with low creditor guarantee ($\gamma=0.2$). This can be partly explained by the fact that the return on the marginal loan is -13.6% with high guarantees versus -2.3% with low guarantees; in both cases, the bank overlends by making net present value (NPV) negative loans at the margin, which arise due to various distortions in the model (e.g., limited liability, the benefit of cheap debt), but the magnitude is considerably greater in the case of high guarantees.

1.4 Testable predictions

The model makes two key testable predictions. The first, which has been previously emphasized, is that net equity issuance is countercyclical when government guarantees to creditors are strong and procyclical when government guarantees to creditors are weak or absent. This prediction is clear from the numerical results in Figures 1 and 2. The second is that bank funding costs are sensitive to equity issuance and payout announcements when government guarantees to creditors are weak or absent but that this sensitivity is dulled in the presence of strong creditor guarantees. This prediction relates to a key mechanism of the model—"creditor market discipline"—which can counteract risk-shifting by bankers and is key to explaining the differential patterns of equity issuance and payouts in situations with different levels of creditor guarantees. I test these two predictions in the following sections.

2. Data and Summary Statistics

I construct several data sets: (1) a panel data set with quarterly observations of aggregate credit expansion, assets, equity issuance, payouts, and net income for U.S. publicly traded banks over the sample period 1980–2012 and (2) an analogous data set covering the eleven German Landesbanken from 2000 to 2012, with additional data on subordinated debt yields, equity issuance and payout announcements, and the fraction of guaranteed liabilities in each year. I describe these variables in turn below before providing some summary statistics on the equity issuance and payouts of the twenty largest U.S. commercial banks.

2.1 Data on equity issuance and payouts for U.S. commercial banks

Commercial banks (which I take to also include savings and trust institutions) are selected from Compustat by SIC codes 6020–6036, then further filtered by NAICS and GIC codes to remove erroneous firms; holding companies that are primarily banks (e.g., Citigroup) with SIC codes 6712 are also included as commercial banks. For each quarter, I categorize commercial banks by size: the largest 1–5 banks by assets, 6–20, 21–100, 101–500, and >500 by assets.

The main variable analyzed in this paper is *net equity issuance minus dividends* (new equity issuance minus share repurchases minus dividends).

I first construct *net equity issuance* by looking at the quarterly net change in common shares outstanding from Compustat (after adjusting for stock splits and effective dilutions) and value the equity based on the end-of-quarter price for the equity. Stephens and Weisbach (1998) and Jagannathan, Stephens, and Weisbach (2000) suggest that this method is generally accurate in measuring net common equity issuance.¹⁰ Also following Jagannathan et al. (2000), I decompose *net equity issuance* into *new equity issuance* = $\max(\text{net equity issuance}, 0)$ and *share repurchases* = $\min(\text{net equity issuance}, 0)$. As recommended by Jagannathan et al. (2000), this decomposition is first done on the *monthly* level for each bank using monthly CRSP data, to minimize netting, then aggregated to quarterly observations. Jagannathan et al. (2000) shows this method is generally accurate compared to other methods in measuring equity issuances and repurchases. I also gather data on *dividends*, *assets*, *book equity*, and *net income*. The variables are extracted at the individual bank level at a quarterly frequency, then normalized by beginning-of-quarter *book equity*.¹¹

As the key variable representing the credit cycle, I define the variable *credit expansion*, also denoted $\Delta(\text{bank credit} / \text{GDP})_t$, as the year-over-year change in aggregate bank credit to gross domestic product (GDP), where bank credit is defined as credit from banks to domestic households and private nonfinancial corporations and taken from the BIS long series on domestic bank credit. I then standardize *credit expansion* by subtracting out the country's mean and dividing by the standard deviation. This paper analyzes cyclicity in terms of *credit expansion*, because credit expansions are when the asset side of banks' balance sheets is expanding. This paper thus looks at how those assets expansions are financed between debt and equity and finds that banks actually decrease their rate of equity growth during asset expansions. Nevertheless, similar results hold when defining cyclicity in terms of GDP or other variables.

To analyze the reaction of U.S. bank creditors to equity issuance and payout announcements, I use Capital IQ to gather the announcement dates of common equity issuance and repurchases, and CRSP to gather the announcement dates of common equity dividend increases or decreases. This sample of announcement events covers the twenty largest U.S. commercial banks in each year over the

¹⁰ All equity variables (e.g., net issuance, dividends, book equity) in this paper solely refer to common equity.

¹¹ All firm-quarters featuring "large" mergers and acquisitions are excluded, where "large" is defined as growth in excess of 10% in either assets or book equity of the acquirer in a single quarter. Excluding mergers is critical, because the majority of bank equity issuance in the data is from mergers and acquisitions using stock as the deal currency. However, this new issuance is illusory, as there is actually no new net equity created in the aggregate banking sector. The reason is that an acquirer issuing new stock to exchange for a target company's stock does not create new equity in the aggregate but simply substitutes one company's share (the target's) for another (the acquirer's). Thus, these events are excluded from the sample. To do so, I systematically identify all "large" mergers and acquisitions (M&As) in the sample. I start by selecting out all firm-quarters in which growth in either assets or book equity is in excess of 10% in the quarter, which often indicates a potential M&A. Then, going through each selected firm-quarter individually, I investigate news accounts, Capital IQ, and Moody's Manuals on Banking to determine which ones actually are M&As.

period 1996–2012. For each of these announcement events, I also use Capital IQ to gather data on unsecured subordinated debt yields to assess the change in yields around each announcement event.

2.2 German Landesbank data

For each of the eleven German Landesbanken in the sample, I collect annual data from 2000 to 2012 on *common equity issuance, repurchases, dividends, net income, changes in retained income, assets, and book equity*. The main data source is Capital IQ, which is supplemented with Bankscope and individual bank financial statements in cases in which data are missing.¹²

For each Landesbank, I use Capital IQ to gather the announcement dates of common equity issuance, repurchases, and dividend increases or decreases. Online Appendix Table 7 lists the announcement dates. For each of these announcement events, I also use Capital IQ to gather data on unsecured subordinated debt yields. Given that there are usually various bond issues outstanding around each announcement, I choose corresponding bond issues so that the guaranteed and nonguaranteed bonds have a close match in terms of their seniority, modified duration, and other characteristics. The matching procedure is detailed in Online Appendix Section C, with bond characteristics reported in Online Appendix Table 7.

For each year and Landesbank, I also calculate the fraction of debt that is guaranteed. As discussed in detail in Section 4.1, the 2001 “Brussels Agreement” phased out the guarantees according to the following rules: liabilities issued between July 19, 2001, and July 18, 2005, and maturing no later than December 31, 2015, were still covered by the guarantor’s liability, while those issued until July 18, 2005, but maturing after December 31, 2015, were not guaranteed; liabilities issued before the July 18, 2001, agreement were grandfathered in and maintained their guarantees, regardless of their maturities. Using these rules, I also calculate the fraction of debt that is guaranteed for each bank-year by first gathering information from Capital IQ on all outstanding debt issues for each Landesbank over the period 1996–2012, then using information on the bond’s issue date and maturity to classify it as guaranteed or nonguaranteed, and lastly aggregating the outstanding value of guaranteed versus nonguaranteed debt in each year to compute the fraction of guaranteed-to-total debt for each bank and each year.

2.3 Summary statistics

Table 1 presents summary statistics related to equity issuance, payouts, and other key variables for large U.S. banks. Panel A presents cross-sectional

¹² The eleven Landesbanken during this period are Bayern LB, Bremer LB, HSH Nordbank (formed from the 2003 merger of Hamburgischen LB and LB Schleswig-Holstein), LaBa Berlin, LB Baden-Württemberg (LBBW), LB Hessen-Thüringen (Helaba), LB Rheinland-Pfalz (LRP, acquired by LBBW in 2005), LB Saar, Norddeutsche LB, Sachsen LB, and West LB.

Table 1
Summary statistics

A. Cross-sectional statistics, 20 largest banks at the end of 2005

	Mean	SD	5%	10%	25%	50%	75%	90%	95%
<i>Bank balance sheet statistics</i>									
Assets (\$ millions)	341,787	460,108	53,414	55,146	91,954	109,170	481,741	1,291,803	1,494,037
Book equity / Assets	9.2%	2.1%	6.5%	6.7%	7.8%	9.0%	9.7%	12.5%	15.6%
Growth of assets (annualized)	8.0%	5.6%	0.7%	0.8%	3.1%	7.7%	12.9%	16.3%	16.6%
<i>Equity flows (as a percentage of book equity, annualized)</i>									
Net equity issuance minus dividends	-12.5%	6.7%	-26.4%	-22.2%	-16.9%	-10.9%	-7.6%	-4.3%	-2.9%
Equity issuance	0.5%	1.3%	0.0%	0.0%	0.0%	0.0%	0.1%	2.1%	5.0%
Repurchases	5.9%	5.7%	0.0%	0.0%	1.6%	3.0%	10.6%	14.1%	19.0%
Dividends	6.8%	2.6%	1.4%	3.5%	4.7%	7.3%	8.1%	9.5%	12.0%
Net income minus payouts	3.4%	6.8%	-13.7%	-3.9%	0.6%	4.0%	6.0%	11.2%	17.6%
Change in book equity	5.3%	7.7%	-9.6%	-2.5%	1.6%	4.2%	10.1%	17.1%	22.0%

B. Time series statistics, quarterly observations (annualized) over 1980-2012

	Mean	SD	5%	10%	25%	50%	75%	90%	95%
<i>Credit expansion</i>									
Δ (Bank credit / GDP)	-0.1%	2.0%	-4.3%	-3.3%	-1.4%	0.3%	1.5%	2.2%	2.4%
<i>Equity flows (aggregated across the 20 largest banks by assets, expressed as a percentage of book equity)</i>									
Net equity issuance minus dividends	-3.2%	7.9%	-13.5%	-10.9%	-7.4%	-3.5%	-1.0%	2.9%	8.5%
Equity issuance	3.5%	3.9%	0.2%	0.5%	1.0%	2.2%	5.0%	8.3%	9.8%
Repurchases	2.2%	3.0%	0.0%	0.0%	0.0%	0.9%	3.3%	5.9%	8.6%
Dividends	5.1%	2.1%	0.9%	1.7%	4.4%	5.4%	6.3%	7.3%	7.8%
Net income minus payouts	5.1%	9.4%	-5.7%	-0.5%	3.8%	6.5%	9.6%	12.0%	14.4%
Change in book equity	7.8%	11.2%	-5.5%	-0.2%	3.4%	8.4%	13.2%	16.7%	18.8%

This table presents summary statistics for equity issuance, dividends, repurchases, and other key variables related to the common equity of the 20 largest U.S. commercial banks by assets. Panel A presents time-series statistics aggregated across banks over the period 1980-2012. Panel B presents cross-sectional statistics for the twenty largest banks at the end of 2005, around the peak of the U.S. credit cycle.

statistics for the twenty largest banks during 2005, around the peak of the most recent credit cycle. Summary statistics are reported for *bank assets*, *book equity to assets*, *annualized growth of assets*, and *annualized growth of book equity*. This panel demonstrates substantial heterogeneity across large banks, even at the peak of the credit cycle. For example, there is substantial variation in leverage across large banks, from 6.5% to 15.6% book equity to assets. There is also substantial heterogeneity in individual banks' balance sheet expansion, with growth of assets ranging from 0.7% to 16.6% and growth of book equity from -9.6% to 22%; thus, even though all firms are increasing their assets (with a mean growth rate of 8.0%), some banks are decreasing their book equity (which is not due to losses, because net income is positive for all these banks in 2005).

Panel A also shows substantial heterogeneity in equity issuance and payouts in the cross-section of large banks in 2005. Summary statistics are reported for *net equity issuance minus dividends*, *issuance*, *repurchases*, *dividends*, and *net income minus payouts* (i.e., changes in retained earnings), all expressed as a percentage of book equity. *Net equity issuance minus dividends* has a mean of -12.5%, suggesting that banks choose to pay out high levels of capital during credit expansions. *New issuance* and *share repurchases* are 0% of book equity for about half of banks, but range as high as 5.0% (for issuance) and 19.0% (for repurchases). *Dividend payouts* range from 1.4% to 12.0%.

Panel B presents aggregated time-series statistics over the period 1980–2012. The first row looks at *credit expansion* (the year-over-year change in bank credit to GDP). Figure 3, panel A, plots *credit expansion*, along with alternative measures of the credit cycle, such as the change in total credit to GDP (from the BIS) and the change in bank loans to GDP (from the Fed flow of funds), which are shown to be similar. As shown in Table 1, panel B, *credit expansion* can range as high as 2.4% of GDP (i.e., a credit expansion) and as low as -4.3% (i.e., a credit contraction).

Looking at equity issuance and payouts over time for the twenty largest banks aggregated together, Table 1, panel B, reports that *net equity issuance minus dividends*, its components (*issuance*, *repurchases*, *dividends*), and *net income minus payouts* (i.e., changes in retained earnings) are highly time-varying over credit cycles, ranging from -13.5% to 8.5% for equity *net issuance minus dividends*, and from -5.7% to 14.4% for *net income minus payouts*. Figure 3, panel B, plots *net equity issuance*, *dividends*, and *net income minus payouts* (i.e., changes in retained earnings) over time for the twenty largest banks aggregated, which are shown to be highly cyclical. As shown later in the paper, large banks' net equity issuance and changes in retained earnings are countercyclical over the credit cycle over the period 1980–2012, that is, net payouts during credit expansions, followed by net recapitalization during credit contractions.

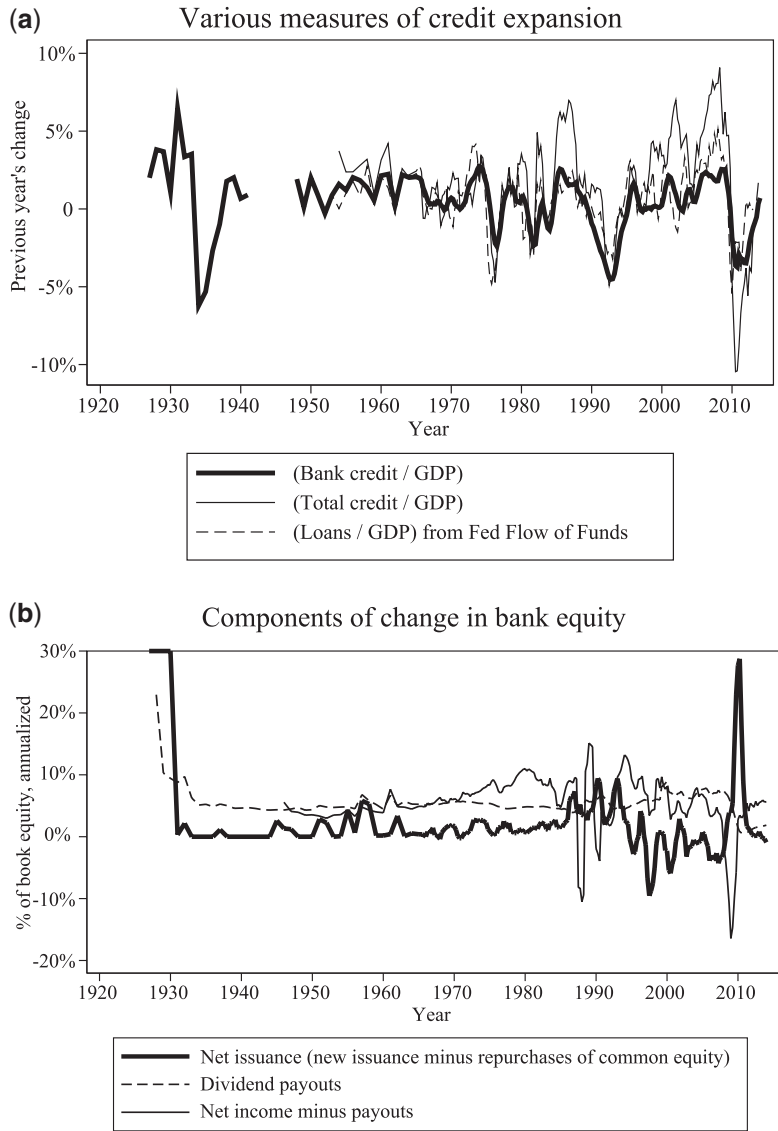


Figure 3
Credit expansion and bank equity
Panel A plots bank credit expansion (defined as the previous year's change in the ratio of bank credit to GDP) over time, along with alternative measures of the credit cycle, such as the change in total credit to GDP and change in aggregate bank loans to GDP. Panel B plots net issuance of common equity, dividends, and net income minus payouts (as a percentage of book equity), aggregated over the twenty largest U.S. commercial banks.

3. Patterns of Bank Equity Issuance and Payouts

In this section, I present the first main empirical result of this paper, that net equity issuance by U.S. large banks over the period 1980–2012 is countercyclical. Then, sorting banks by size, I show that there is a robust size factor: the largest banks are the most strongly countercyclical, while smaller banks show a less countercyclical or acyclical pattern—and that these patterns are sizeable in economic magnitudes. Then I explore various hypotheses that might explain this size factor, including the potential role of government guarantees to creditors, using data on Fitch support ratings.

3.1 Countercyclical equity issuance for large U.S. banks

To analyze the cyclicity of equity issuance, payouts, and retained earnings, I estimate the following bank-level panel regression with quarterly observations and bank fixed effects:

$$y_{i,t} = \alpha_i + \Delta(\text{bank credit}/\text{GDP})_t (\beta_1 + \beta_2' X_{i,t}) + \gamma' \text{Controls}_{i,t} + \varepsilon_{i,t}. \quad (1)$$

The standard dependent variable $y_{i,t}$ is *net equity issuance minus dividends* for each bank, though I also look at other dependent variables, such as its components (*new issuance*, *repurchases*, *dividends*), *additions to retained earnings*, and *total change in book equity*. *Credit expansion*, that is, $\Delta(\text{bank credit}/\text{GDP})_t$, the standardized year-over-year change in aggregate bank credit to GDP, is the standard measure of the credit cycle. The left-hand side variable (*net equity issuance minus dividends*) and the right-hand side variable (*credit expansion*) are chosen to be *contemporaneous* because this regression asks, when the asset side of banks' balance sheets expands (*credit expansion*), how the asset expansions are financed between debt and equity. Equation (1) thus tests whether large banks decrease their rate of equity growth during these asset expansions.^{13, 14}

The coefficient β_1 measures the overall cyclicity of bank equity issuance. In some specifications, I analyze the cyclicity for banks of various size, by interacting $\Delta(\text{bank credit}/\text{GDP})_t$ with indicator variables for bank sizes, specifically by letting $X_{i,t}$ be a vector of the following indicator variables: $1[\#1-5 \text{ by assets}]_{i,t}$, $1[\#6-20 \text{ by assets}]_{i,t}$, $1[\#21-100 \text{ by assets}]_{i,t}$, $1[\#101-500 \text{ by assets}]_{i,t}$, and $1[\#501+ \text{ by assets}]_{i,t}$. In specifications with these interactions, the bank size indicator variables are also included outside the interaction term

¹³ Another reason for using $\Delta(\text{bank credit}/\text{GDP})$ as the measure of cyclicity is that previous studies (e.g., Schularick and Taylor 2012) demonstrate this variable to be a robust predictor of financial crises, and, thus, for financial stability reasons, it is important to ask whether banks raise more equity when $\Delta(\text{bank credit}/\text{GDP})$ is increasing.

¹⁴ Note that the cyclicity of $\Delta(\text{bank credit}/\text{GDP})$ is mainly driven by the cyclicity of the numerator (bank credit) rather than that of the denominator (GDP), as shown in Baron and Xiong (2017). Although the results similarly hold using $\Delta(\text{bank credit})$, the variable $\Delta(\text{bank credit}/\text{GDP})$ is standard in the literature on predicting financial distress and hence why it is used here.

to fully saturate the model, though the results are nearly identical with and without these terms outside the interaction. Various bank-level control variables are employed. The first, $(Income/book\ equity)_{i,t}$, is the previous year's bank net income (normalized by the previous year's book equity), which controls for the fact that equity issuance and payouts may be driven in part by time-varying cash-flows. Later in this subsection, I also demonstrate in various other ways that countercyclical issuance is not simply accounted for by the increased profitability of banks during credit expansions. The second, $(Market/book)_{i,t}$, the bank market-to-book ratio, controls for the fact that issuance and payouts may be driven by either market timing concerns or growth opportunities reflected by this ratio. The third, $(Past\ year\ stock\ return)_{i,t}$, the previous year's total return of the bank's stock price, accounts for potential manager market timing. The fourth is $1[Undercapitalized]_{i,t}$, an indicator variable that equals 1 (and 0 otherwise) when the previous year's common equity to assets ratio is less than 5%, as banks may be forced to recapitalize by regulators when their equity falls below minimum capital requirements. In Section 3.4, I add other variables to test a host of alternative hypotheses that may in part drive patterns of bank equity issuance and payouts. The control variables described above are simply the baseline controls.

The panel regression is estimated on quarterly data with standard errors doubly clustered on bank and time. The results are reported in Table 2, panel A, with *net equity issuance minus dividends* as the dependent variable. Column 1 estimates the bank-level panel regression (Equation (1)) in a univariate framework, with just $\Delta(bank\ credit/GDP)_t$ as the independent variable. A 1-standard-deviation increase in *credit expansion* is associated with a statistically significant 1.2-percentage-point decrease in *net equity issuance minus dividends* ($t = -5.492$). Column 2 adds in the above controls, but the coefficient on *credit expansion* remains unchanged. Note that the coefficients on the control variables are significant and all in the expected direction: an increase in net income is associated with greater net payouts, while higher market-to-book, higher past year stock returns, and being undercapitalized are associated with increased equity issuance, as expected. Thus, during credit expansions, banks issue less new equity and pay out more to shareholders, even controlling for firm income and other variables.

The coefficients reported in Columns 1 and 2 are an average across all U.S. commercial banks in the sample, which may mask heterogeneity across banks. Therefore, Columns 3 and 4 report estimates with *credit expansion* interacted with indicators of bank size, $1[\#1-5\ by\ assets]_{i,t}$ through $1[\#501+ \ by\ assets]_{i,t}$. Looking at the estimates in Column 3, the cyclicity coefficient for banks #1–5 by assets is even more negative in magnitude (-0.030 , with a t -statistic of -3.157 , meaning that a 1-standard-deviation increase in *credit expansion* is associated with a statistically significant 3.0-percentage-point decrease in *net equity issuance minus dividends*). Comparing the coefficients in rows 2

Table 2

A. Net equity issuance minus dividends over the credit cycle for banks of various sizes

	Net equity issuance minus dividends			
	(1)	(2)	(3)	(4)
Δ (Bank credit / GDP) _t	-0.012***	-0.012***		
	[-5.492]	[-5.095]		
Δ (Bank credit / GDP) _t				
× 1[#1-5 by assets] _{i,t}			-0.030***	-0.030***
			[-3.157]	[-2.988]
× 1[#6-20 by assets] _{i,t}			-0.019***	-0.021***
			[-3.110]	[-3.281]
× 1[#21-100 by assets] _{i,t}			-0.014**	-0.014**
			[-2.397]	[-2.366]
× 1[#101-500 by assets] _{i,t}			-0.013***	-0.014***
			[-6.077]	[-5.809]
× 1[#501+ by assets] _{i,t}			-0.001	-0.000
			[-0.344]	[-0.029]
(Income / book equity) _{i,t}		-0.011***		-0.012***
		[-3.239]		[-3.482]
(Market / book) _{i,t}		0.013***		0.015***
		[3.430]		[3.855]
(Previous year's stock return) _{i,t}		0.039***		0.039***
		[6.136]		[6.040]
1[Undercapitalized] _{i,t}		0.057***		0.054***
		[7.423]		[7.011]
1[#1-5 by assets] _{i,t}			-0.115***	-0.109***
			[-4.860]	[-4.567]
1[#6-20 by assets] _{i,t}			-0.077***	-0.077***
			[-4.164]	[-3.879]
1[#21-100 by assets] _{i,t}			0.003	0.003
			[0.203]	[0.252]
1[#101-500 by assets] _{i,t}			0.010	0.012*
			[1.583]	[1.781]
Constant	0.017***	0.012***	0.013**	0.008
	[7.687]	[6.633]	[2.426]	[1.302]
<i>Difference in cyclical coeff.:</i>				
#1-5 minus #501+			-0.029***	-0.030***
			[-2.744]	[-2.795]
#6-20 minus #501+			-0.018**	-0.021***
			[-2.521]	[-2.954]
#21-100 minus #501+			-0.013*	-0.014**
			[-1.875]	[-2.056]
#101-500 minus #501+			-0.012***	-0.013***
			[-3.821]	[-4.460]
Bank fixed effects	Yes	Yes	Yes	Yes
Adj. R ²	.04	.05	.04	.05
N	64,955	63,978	64,955	63,978

(Continued.)

through 6, the magnitudes of the coefficients are monotonically decreasing for banks of smaller size: for banks of size #501+, the coefficient is statistically indistinguishable from zero. Column 4 shows that these results are nearly unchanged in the presence of control variables. At the bottom of the table, differences between coefficients on banks of various sizes (relative to banks of size #501+) are tested and found to be statistically significant in all cases. Thus, I conclude that the largest banks have strongly countercyclical patterns

Table 2
Continued

B. Equity issuance, payouts, and changes in retained earnings over the credit cycle

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Equity issuance		(negative) Repurchases		(negative) Dividends		Net income minus payouts		Δ (Book equity)	
Δ (Bank credit / GDP) _{<i>t</i>}	-0.007*** [-3.500]		-0.003*** [-4.712]		-0.001*** [-4.310]		-0.009*** [-3.162]		-0.010* [-1.908]	
Δ (Bank credit / GDP) _{<i>t</i>} × 1[#1-5 by assets] _{<i>t,t</i>}		-0.017** [-2.091]		-0.007*** [-3.015]		-0.006*** [-2.720]		-0.019*** [-2.639]		-0.031*** [-2.639]
× 1[#6-20 by assets] _{<i>t,t</i>}		-0.011** [-2.156]		-0.007*** [-3.876]		-0.003** [-2.821]		-0.016*** [-2.917]		-0.023*** [-2.917]
× 1[#21-100 by assets] _{<i>t,t</i>}		-0.008 [-1.402]		-0.004** [-4.512]		-0.002*** [-4.269]		-0.012*** [-4.595]		-0.012*** [-4.595]
× 1[#101-500 by assets] _{<i>t,t</i>}		-0.009*** [-4.016]		-0.003*** [-3.822]		-0.001*** [-3.682]		-0.009*** [-3.682]		-0.011* [-1.870]
× 1[#501+ by assets] _{<i>t,t</i>}		0.002 [1.025]		0.002* [-1.780]		-0.001 [-1.326]		-0.004 [-1.196]		0.001 [0.099]
(Income / book equity) _{<i>t,t</i>}	-0.007** [-1.971]	-0.007** [-2.132]	-0.000 [-0.870]	-0.001 [-1.173]	-0.004*** [-12.321]	-0.004*** [-12.321]	0.138*** [38.696]	0.138*** [38.696]	0.065*** [8.070]	0.064*** [7.997]
(Market / book) _{<i>t,t</i>}	0.027*** [7.962]	0.029*** [8.426]	-0.006*** [-6.178]	-0.006*** [-5.999]	-0.008*** [-14.866]	-0.008*** [-14.570]	-0.017*** [-8.311]	-0.016*** [-8.142]	0.024*** [4.120]	0.026*** [4.120]
(Previous year's stock return) _{<i>t,t</i>}	0.035*** [5.970]	0.035*** [5.995]	-0.000 [-0.033]	-0.000 [-0.143]	0.004*** [5.825]	0.004*** [5.331]	0.022*** [3.481]	0.021*** [3.350]	0.057*** [4.717]	0.055*** [4.595]
1[Undercapitalized] _{<i>t,t</i>}	0.050*** [6.454]	0.047*** [6.106]	0.005*** [3.664]	0.005*** [3.455]	0.002*** [2.592]	0.002*** [2.815]	0.025*** [4.171]	0.025*** [4.141]	0.149*** [9.551]	0.146*** [9.551]
1[#1-5 by assets] _{<i>t,t</i>}		-0.098*** [-4.566]		-0.013* [-1.683]		0.002 [0.521]		-0.009 [-0.536]		-0.109*** [-3.816]
1[#6-20 by assets] _{<i>t,t</i>}		-0.061*** [-3.317]		-0.011** [-2.205]		-0.005* [-1.789]		-0.023** [-2.567]		-0.089*** [-4.210]
1[#21-100 by assets] _{<i>t,t</i>}		0.018 [1.422]		-0.007*** [-2.617]		-0.007*** [-5.070]		-0.027*** [-5.303]		-0.022 [-1.610]
1[#101-500 by assets] _{<i>t,t</i>}		0.016** [2.397]		0.000 [0.021]		-0.004*** [-4.854]		-0.013*** [-4.102]		-0.003 [-0.303]
Constant	0.067*** [45.260]	0.058*** [10.177]	-0.021*** [-26.928]	-0.020*** [-13.650]	-0.033*** [-126.707]	-0.030*** [-43.061]	0.020*** [10.426]	0.032*** [11.631]	0.081*** [17.128]	0.089*** [10.289]
<i>Difference in cyclical coeff:</i>										
#1-5 minus #501+		-0.020** [-2.218]		-0.005** [-2.052]		-0.006** [-2.277]		-0.015** [-2.211]		-0.032** [-2.314]
#6-20 minus #501+		-0.014** [-2.346]		-0.005** [-2.424]		-0.003* [-1.914]		-0.012* [-1.914]		-0.023** [-2.409]
#21-100 minus #501+		-0.010 [-1.644]		-0.002* [-1.666]		-0.001** [-2.291]		-0.008** [-2.468]		-0.012 [-1.541]
#101-500 minus #501+		-0.011*** [-4.370]		-0.001 [-1.222]		-0.001** [-2.038]		-0.005*** [-3.095]		-0.012** [-2.718]
Bank fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R ²	.04	.04	.12	.12	.65	.65	.51	.51	.08	.09
N	63,978	63,978	63,978	63,978	63,978	63,978	63,957	63,957	63,978	63,978

This table demonstrates that equity issuance, payouts, and changes in retained earnings are countercyclical across credit cycles over the period 1980-2012 for the largest U.S. commercial banks. The table reports panel regression estimates from Equation (1). In panel A, the dependent variable is net issuance of common equity (Columns 5 and 6), dividends to common equity (Columns 5 and 6), repurchases of common equity (Columns 5 and 6), dividends to common equity (Columns 5 and 6), net income minus dividends and repurchases, Columns 7 and 8), and change in book equity (Columns 9 and 10). The dependent variables are all expressed as fractions of book equity. The main independent variable (row 1) is Δ (Bank credit / GDP), the previous year's net income (as a fraction of the previous year's book equity); (Market / book)_{*t,t*}, the beginning-of-quarter market-to-book ratio; (Previous year's stock return)_{*t,t*}, the previous year's total return on the bank's stock; and 1[Undercapitalized]_{*t,t*}, an indicator variable that equals 1 when the previous year's equity-to-assets ratio of a bank is less than 5%. All continuous independent variables are expressed in units of standard deviations. At the bottom of each panel, the observations are double-clustered on firm and quarter. * $p < .1$; ** $p < .05$; *** $p < .01$.

of net equity issuance, while smaller banks have little or no cyclical in their patterns of equity issuance.^{15, 16, 17}

It is important to note that this pattern of countercyclical equity issuance is not only driven by banks recapitalizing—potentially involuntarily—during credit contractions but also by banks issuing less equity during credit expansions. To examine this issue, Online Appendix Table 1 reports estimates from a specification that interacts $\Delta(\text{bank credit}/\text{GDP})_t$ with indicator variables positive_t (which equals 1 if $\Delta(\text{bank credit}/\text{GDP})_t \geq 0$), and negative_t (which equals 1 if $\Delta(\text{bank credit}/\text{GDP})_t < 0$). These panel results show that banks both issue less equity during credit expansions and subsequently recapitalize during credit contractions. If anything, the countercyclical pattern is slightly stronger during times of positive credit expansion. This result holds both for the overall banking sector (Columns 1 and 2) and for banks decomposed by size (Columns 3 and 4). Section 3.2 and Table 3 present further evidence of this conclusion. And even if the pattern were only driven by banks recapitalizing during the bust, the fundamental question would still remain: if everyone understands that the government or market will force costly equity issuance in bad times, why not issue equity in advance, in good times?

Returning to Table 2, in panel B, I also look at other measures of equity inflows and outflows of large commercial banks: *new issuance* (Columns 1 and 2), *repurchases* (Columns 3 and 4), *dividends* (Columns 5 and 6), *net income minus payouts* (i.e., changes in retained earnings, Columns 7 and 8), and *total change in book equity* (the quarterly change in reported common book equity, Columns 9 and 10). I repeat my analysis with these as alternative dependent variables, all of which are normalized by book equity. As in panel A, I estimate regressions with and without controls and also interact *credit expansion* with bank size indicator variables. I find that all these components of equity flows are similarly countercyclical, indicating that countercyclical equity issuance is not driven in particular by any one of them, although *new issuance* generally has a somewhat larger magnitude than dividends and repurchases.

¹⁵ These results also can be visualized in the aggregate. Online Appendix Figure 1 plots quarterly observations of *net equity issuance minus dividends*, aggregated over the twenty largest U.S. commercial banks by assets, as a function of *credit expansion*. The pattern is strongly countercyclical. (Methodology and results are discussed in Online Appendix Section B.5.) Online Appendix Figure 2, panel A, plots the same but for aggregated banks of size #21–200, #101–500, and #501+ by assets. As with the panel regression results, these plots show that smaller banks in the aggregate demonstrate little or no cyclical in their patterns of equity issuance.

¹⁶ These results are robust to various alternative specifications, measures of equity issuance and repurchases, and measures of credit cycles. See the Online Appendix Section B.3.

¹⁷ Note there is a large increase in the number of banks in the data set starting in 1994, when there is a jump from ~200 banks to ~800 banks. I therefore conduct robustness analysis to ensure that this increase in the number of banks in 1994 in no way drives the results. First, I show in Online Appendix Table 3 that the results of the paper are robust to estimating them on the subsample 1994–2012, in other words, when the group [#501+ by assets] is nonempty. In fact, the results are even quantitatively stronger over this subsample. Second, I show in Online Appendix Table 4 that the results of the paper are robust to combining the bank size indicators 1[#101–500 by assets] and 1[#501+ by assets] into a single indicator 1[#101+ by assets], so that the smallest bank indicator is not empty in the pre-1994 period. Again, the results are qualitatively unchanged.

One potential concern might be that equity issuance and payouts may simply be driven by cash flow considerations: during credit expansions, banks might be highly profitable and generate more profit than they have opportunities for reinvestment, leading them to pay out excess profits. If this were the case, one would expect payouts to increase proportionately with income. However, Columns 7 and 8 of Table 2, panel B, show that *net income minus payouts* (i.e., changes in retained earnings) is countercyclical for the largest U.S. commercial banks, meaning that banks increase their payouts faster than their income increases during credit expansions. This implies that although bank profitability is higher during credit expansions, the largest banks choose to increase their payouts *even more*, to more than offset their increased profitability. The largest banks thus actively choose to reduce their equity by increasing payouts during credit expansions, despite retained earnings being a potentially efficient way to increase equity.¹⁸

From this subsection, I conclude that equity issuance of U.S. commercial banks is countercyclical across credit cycles and, most importantly, that there is a strong and robust size factor: the largest banks are the most strongly countercyclical, whereas smaller banks show a less countercyclical or acyclical pattern. In Section 3.4, I explore various hypotheses that might explain this size factor, including the potential role of government guarantees to creditors.

3.2 Magnitudes

Next, I study the magnitudes of bank equity issuance and payouts relative to subsequent banking losses. I find in this subsection that the “equity gap” (defined as the reduced equity issuance relative to trend) and payouts are large in magnitude relative to subsequent banking losses.

Table 3 reports the aggregate magnitudes (in millions of dollars, not inflation adjusted) of bank equity net issuance (relative to trend) and payouts across selected credit expansions and contractions in the United States from 1980 to 2012.¹⁹ The sample of banks consists of the twenty largest U.S. commercial banks at each point in time. It is easy to verify that equity issuance minus payouts is indeed countercyclical in the aggregate, because the equity issuance magnitudes are smaller during the expansions than the contractions (and vice versa for payouts), clearly demonstrating the robustness of the countercyclical pattern. In particular, this alternating negative and positive sequence of net equity issuance across recent cycles demonstrates that the countercyclical pattern is not simply driven by banks recapitalizing in downturns.

Most importantly, the “equity gap” is large in magnitude relative to subsequent banking losses. For example, in the most recent credit expansion

¹⁸ The countercyclicality of retained earnings is not driven by increased provisioning during credit expansions, because the measure of retained earnings used here is *before* changes to provisions and loan-loss reserves.

¹⁹ For the purpose of calculating the “equity gap,” the trend is calculated by plotting equity issuance over time and estimating a line of best fit at each point in time using a ± 10 -year moving window.

Table 3
Magnitudes of bank equity issuance and payouts relative to subsequent bank losses
20 largest U.S. commercial banks, all quantities in units of million \$

	Start	End	Δ (Bank credit / GDP)	Book eq., start year	Net equity issuance (rel. trend)	Dividends	Issuance	Repurch.	Net income	Bank losses
1	Credit expansion 1979q2	1980q1	0.78%	23,765	651	4,675	651	0	14,907	
	Contraction 1980q2	1981q4	-1.70%	26,673	3,638	9,431	3,704	66	28,740	-769
2	Credit expansion 1982q1	1982q3	0.63%	32,310	1,090	4,641	1,123	33	12,605	
	Contraction 1982q4	1984q1	-0.89%	34,462	5,004	9,983	5,349	346	26,698	-10,523
3	Credit expansion 1984q2	1986q4	1.98%	33,594	17,780	18,573	19,081	1,301	54,673	
	Contraction 1989q1	1993q4	-2.69%	57,433	72,067	72,552	75,021	2,955	159,064	-51,938
4	Credit expansion 1995q1	1997q4	0.55%	111,683	-59,061	99,532	46,230	105,291	319,596	
	Contraction 1998q1	1999q2	0.09%	163,697	9,155	73,103	49,152	39,996	204,096	-16,045
5	Credit expansion 1999q3	2000q3	1.53%	248,907	-63,663	88,535	28,434	92,097	225,159	
	Contraction 2000q4	2001q4	1.14%	176,861	18,036	89,887	55,993	37,957	179,599	-8,808
6	Credit expansion 2005q1	2007q3	2.05%	562,780	-211,355	451,321	95,603	306,958	977,003	
	Contraction 2007q4	2012q1	-1.26%	701,697	938,149	301,136	1,068,756	130,607	417,316	-538,084

This table reports the aggregate magnitudes (in millions of dollars, not inflation adjusted) of bank equity net issuance (relative to trend) and other related quantities across credit expansions and contractions in the United States from 1980 to 2012. The sample of banks consists of the twenty largest U.S. commercial banks at each point in time. The variable Δ (Bank credit / GDP) is measured in this table as the *cumulative* change in bank credit over the entire credit expansion or contraction (as a fraction of GDP at the start of the expansion). Bank losses are computed as described in the text.

(2004–2007), the decreased net equity issuance (relative to trend) was more than \$200 billion and dividend payouts were in excess of \$450 billion for the twenty largest U.S. banks, compared to roughly \$500 billion in subsequent losses for these banks during the 2007–2008 financial crisis. Losses are estimated by adding up the quarterly net income of banks conditional on being negative for a given bank-quarter. Thus, in the counterfactual in which banks substantially reduce payouts and do not reduce new issuance during the boom, banks would have had more than sufficient buffer to withstand subsequent credit losses.

3.3 Small banks raise equity often and in relatively large amounts

It is commonly argued that banks are often resistant to raising new equity and do so rarely (e.g., Admati et al. 2013; Adrian and Shin 2013). However, I show here that, contrary to what might be supposed from the prior literature, smaller banks actually raise equity often and in relatively large amounts. This finding further reinforces the notion that, whatever frictions prevent banks from raising new equity during credit expansions, these frictions are most pronounced for the largest banks.

Table 4 reports the frequency and average amount of new equity issuance by U.S. commercial banks of various sizes over the period 1980–2012. The *frequency* of bank equity issuance is computed as the percentage of bank-quarters in which new equity issuance is greater than 5% (of beginning-of-quarter book equity). The *average amount* of new equity issuance is computed as the average new equity issuance, conditional on bank-quarters in which new equity issuance is greater than 5%. (The greater-than-5% threshold is used to filter noise; the relative patterns reported in Table 4 are robust to other thresholds.)

According to Table 4, banks of size #1–5 by assets issue equity infrequently (1.9% of bank-quarters). However, smaller banks do so more frequently, with the rate monotonically increasing in bank size, up to a maximum of 10.4% of bank-quarters for banks of size #101–200. Beyond that size, however, the frequency of equity issuance begins to decrease, perhaps because the transaction costs involved in issuing new equity become large relative to the size of the bank.

Table 4 also shows a similar pattern for the amount of equity raised, conditional on an equity increase in that bank-quarter: 10.6% of book equity for banks of size #1–5, increasing to a peak of 24.1% of book equity for banks of size #201–500 by assets. The last two rows of the table statistically test the difference between banks of size #101–200 and banks of size #1–5 (or, alternatively, of banks of size #6–20), with *t*-statistics computed from pooled standard errors assuming independent observations. As a robustness test, the right three columns re-perform the calculations during credit expansions only (i.e., when $\Delta(\text{bank credit}/\text{GDP})_t \geq 0$) and find nearly identical results, demonstrating the results are not simply driven by recapitalization during credit contractions.

Table 4
Smaller banks issue new equity often and in larger quantities

	All			Credit expansions only		
	Freq	Avg amount	N	Freq	Avg amount	N
#1-5 by assets	1.9%	10.6%	258	1.6%	13.1%	168
#6-20 by assets	4.7%	12.8%	788	4.3%	12.5%	563
#21-50 by assets	7.6%	10.2%	1,586	7.7%	9.9%	1,131
#51-100 by assets	7.1%	10.5%	2,676	7.2%	10.5%	1,914
#101-200 by assets	10.4%	23.3%	5,527	10.6%	22.5%	3,943
#201-500 by assets	6.8%	24.1%	16,323	6.9%	24.2%	11,640
#501+ by assets	4.9%	22.9%	13,943	4.8%	24.0%	9,662
Diff: (#101-200) minus (#1-5)	8.5%*** [21.5]	12.7%*** [18.5]		9.0%*** [19.1]	9.4%*** [11.7]	
Diff: (#101-200) minus (#6-20)	5.7%*** [15.4]	10.5%*** [15.8]		6.4%*** [14.4]	10.0%*** [12.8]	

This table reports the frequency and average amount of new equity issuance by U.S. commercial banks, broken down by asset size, over the period 1980-2012. The number (“N”) and frequency (“Freq”) of bank equity issuance is defined as the number and percentage, respectively, of bank-quarters in which new equity issuance is greater than 5% of beginning-of-quarter book equity. (The greater-than-5% threshold is used to filter noise; the relative patterns reported in the table are robust to other thresholds.) “Avg amount” is defined as the average amount of each new equity issuance (as a fraction of that bank’s book equity), conditional on bank-quarters in which new equity issuance is greater than 5%. Bank asset size rank (from #1-5 by assets, to #500+ by assets) is assigned each quarter. The last two rows test the difference in the frequency and average amounts of bank equity issuance by size, testing the difference between banks of size #101-200 and banks of size #1-5 (or, alternatively, of banks of size #6-20), with *t*-statistics computed from pooled standard errors assuming independent observations. The left three columns are for the full sample, and the right three columns are for quarters in which Δ (*Bank credit* / *GDP*) > 0.
* *p* < .1; ** *p* < .05; *** *p* < .01.

I thus conclude that small banks raise equity often and in relatively large amounts. Whatever reasons are given to explain why banks resist raising new equity need to explain why these frictions are most pronounced for the largest banks and less evident for small banks. For example, this analysis suggests that transaction costs of equity issuance are unlikely to explain differences in patterns of equity issuance across banks: large banks presumably should have the lowest relative transaction costs, both because fixed costs would be smaller relative to the size of the bank and because the largest banks often have investment banking subsidiaries that can help facilitate equity issuance.

3.4 Evidence on the role of creditor guarantees from Fitch support ratings

Thus far, I have established that bank equity issuance is countercyclical and that there is a robust size factor: the largest banks are the most strongly countercyclical, whereas smaller banks show a less countercyclical or an acyclical pattern. Here, I explore various hypotheses that might explain this size factor, including the potential role of government guarantees to creditors.

To do so, I return to Equation (1) but add in new variables that might indicate which channels might be driving the size factor. Although these results are associational for now, they indicate where to look in subsequent analysis and help disentangle or rule-out other potential hypotheses. One variable that

emerges as a strong contender from this analysis is a bank's Fitch support rating, which measures expectations of government guarantees to creditors. This result motivates the following section, in which I analyze a natural experiment to draw causal inference and explore mechanisms through which government guarantees affect bank equity issuance and payout decisions over the credit cycle.

Fitch support ratings are bank-specific yearly measures of expected government support to banks. Support ratings have thus been frequently used as a proxy for creditor guarantees and bailout probability (e.g., Gropp, Hakenes, and Schnabel 2010; Acharya, Anginer, and Warburton 2016). Fitch's support ratings vary from 1 to 5, with 1 being the highest likelihood of receiving government support, and reflect Fitch's opinion of the likelihood of external support to a bank should it experience financial difficulties. (Importantly, they are *not* a measure for the intrinsic credit quality of a bank.) The Fitch support ratings data are provided by Poghosyan, Werger, and de Haan (2016) in the form of a yearly panel covering 374 U.S. commercial banks over the period 2004 to 2012, which I extend back to 1998 for the twenty largest banks using data from Fitch's Web site. Most U.S. banks, including all banks outside the twenty largest by size, are assigned a rating of 5 for all years (i.e., the lowest likelihood of external support).

I thus reestimate Equation (1), adding in several new independent variables. I first start with an indicator for Fitch support ratings, $1[\text{Fitch support} \leq 3]_{i,t}$, which equals 1 if the bank-year Fitch support rating is less than or equal to 3. I choose a threshold of 3, because support ratings of 4 and 5 correspond to "limited" and "minimal" probabilities, respectively, regarding government support, whereas support ratings of 1 to 3 correspond to a "extremely high" to "moderate" probabilities of external support.

Table 5 reports estimates from Equation (1) over the sample in which Fitch support ratings are available. The dependent variable is *net equity issuance minus dividends*. In Columns 1 and 2, the independent variables are *credit expansion* and *credit expansion* interacted with $1[\text{Fitch support} \leq 3]_{i,t}$. Columns 3 and 4 report a similar specification but one in which *credit expansion* is decomposed by bank size. For comparability, Columns 1 and 3 are estimated on the subsample for which Fitch support ratings data are available. All specifications in Table 5 contain the same bank-level controls as in Table 2.

Column 2 shows that a 1-standard-deviation increase in *credit expansion* is associated with a statistically significant 1.7-percentage-point decrease in *net equity issuance minus dividends* ($t = -6.694$) and an *additional* -3.8-percentage-point decrease ($t = -3.384$) when there is a high expectation of government support (Fitch support rating ≤ 3) for a given bank-year. Column 4 shows similarly, that after decomposing credit expansion by bank size, the coefficient on $1[\text{Fitch support} \leq 3]_{i,t}$ is statistically significant and only

Table 5
Fitch support ratings

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ (Bank credit / GDP) _{<i>it</i>}	-0.018*** [-6.722]	-0.017*** [-6.694]			-0.025*** [-2.626]	0.008 [0.439]		
Δ (Bank credit / GDP) _{<i>it</i>} × 1[Fitch support ≤ 3] _{<i>it</i>}		-0.038*** [-3.384]		-0.027** [-2.174]		-0.024*** [-3.754] -0.036** [-2.464]		-0.052*** [-6.414] -0.037*** [-3.263]
× 1[Multiple branches] _{<i>it</i>}						0.012 [1.604]		0.012 [1.012]
× 1[Foreign branches] _{<i>it</i>}						1.009* [-1.959]		-0.009* [-1.803]
× (RWA / assets) _{<i>it</i>}						0.005 [0.912]		0.003 [0.454]
× (Deposits / debt liabilities) _{<i>it</i>}						-0.007 [-0.955]		-0.002 [-0.258]
× (Securities / assets) _{<i>it</i>}								0.044** [2.057]
× 1[#1-5 by assets] _{<i>it</i>}			-0.049*** [-3.456]	-0.026 [-1.584]			-0.029*** [-3.263]	
× 1[#6-20 by assets] _{<i>it</i>}			-0.034*** [-4.088]	-0.032*** [-3.801]			-0.019* [-1.689]	0.011 [0.691]
× 1[#21-100 by assets] _{<i>it</i>}			-0.032*** [-4.241]	-0.032*** [-4.243]			-0.030*** [-2.910]	0.005 [0.237]
× 1[#101+ by assets] _{<i>it</i>}			-0.015*** [-6.217]	-0.015*** [-6.220]			-0.008 [-0.694]	0.023 [1.522]
Constant	0.008*** [4.299]	0.008*** [4.266]	0.012*** [3.942]	0.012*** [3.949]	0.003 [0.318]	0.008 [0.773]	0.062 [1.538]	0.066 [1.573]
Bank-quarter controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank size indicator controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank fixed effects	.06	.06	.06	.06	.10	.10	.11	.11
Adj. R ²	46.625	46.625	46.625	46.625	2.338	2.338	2.338	2.338
N								

This table is similar to Table 2, panel A, but includes the following additional bank-year variables: 1[Fitch support ≤ 3]_{*it*}, an indicator variable that equals 1 if the bank-year Fitch support rating is ≤ 3; 1[Multiple branches]_{*it*}, an indicator variable that equals 1 if the bank has more than ten branches; 1[Foreign branches]_{*it*}, an indicator variable that takes the value of 1 if the bank had foreign branches; (RWA/assets)_{*it*}, the ratio of risk-weighted assets to assets of a bank; (Deposits/debt liabilities)_{*it*}, the ratio of deposits to total non-equity liabilities; and (Securities/assets)_{*it*}, the ratio of securities holdings to assets of a bank. As in Table 2, panel A, the dependent variable is net issuance of common equity minus dividends, and bank-level control variables (coefficient estimates not reported) are (Income/book equity)_{*it*}, the previous year's net income (as a fraction of the previous year's book equity); (Market/book)_{*it*}, the beginning-of-quarter market-to-book ratio; (Previous year's stock return)_{*it*}, the previous year's total return of the bank's stock price; and 1[Undercapitalized]_{*it*}, an indicator variable that equals 1 when the previous year's equity-to-assets ratio of a bank is less than 5%. Bank size indicator variables (e.g., 1[#1-5 by assets]_{*it*}) outside the interaction term are included (but not reported for space reasons) in the specifications in Columns 3-4 and 7-8. All continuous independent variables are expressed in units of standard deviations. Observations are quarterly. *t*-statistics are computed from standard errors double-clustered on firm and quarter. * $p < .05$; ** $p < .01$; *** $p < .001$.

modestly reduced in magnitude. Note, importantly, that the coefficient on $I[\#1-5 \text{ by assets}]_{i,t}$ is sharply reduced in magnitude and not statistically significant after the addition of $I[\text{Fitch support} \leq 3]_{i,t}$ to the regression, while none of the other size interaction terms for banks of smaller size are affected. This suggests that a large part of the size factor for the top-five largest banks may be due to expectations of government guarantees.²⁰ $I[\text{Fitch support} \leq 3]_{i,t}$ and the size indicators are not collinear: the correlation of $I[\text{Fitch support} \leq 3]_{i,t}$ with $I[\#1-5 \text{ by assets}]_{i,t}$ is 0.73 and with $I[\#1-20 \text{ by assets}]_{i,t}$ is 0.46. Although all banks with Fitch support ratings ≤ 3 are in the top 20 by size, not all banks in the top 20 have Fitch support ratings ≤ 3 (in fact, most do not). There is also variation within banks over time in Fitch support ratings.

Columns 5 to 8 add other variables, all of which are taken from Bankscope and cover a subset of the banks in the main sample over the period 1990–2012. The first of these variables are $I[\text{Multiple branches}]_{i,t}$, an indicator variable that equals 1 if the bank has more than 10 branches, and $I[\text{Foreign branches}]_{i,t}$, an indicator variable that equals 1 if the bank has foreign branches. These variables capture asset diversification across geographical regions, which may affect bank equity issuance and payouts over the cycle. For example, Demsetz and Strahan (1997) find that larger bank holding companies offset their greater asset diversification with lower capital ratios. Other bank-year variables are $(RWA/assets)_{i,t}$, the ratio of risk-weighted assets to assets; $(Securities / assets)_{i,t}$, the ratio of securities holdings to assets, and $(Deposits/debt liabilities)_{i,t}$, the ratio of deposits to total nonequity liabilities. The first of these, $(RWA/assets)_{i,t}$, is a measure of bank risk-taking, as banks with inherently higher risk-tolerances may choose to pay out more equity during credit expansions. The variable $(Securities/assets)_{i,t}$ also serves as a measure of bank risk-taking, as riskier banks generally hold a higher proportion of trading securities; alternatively, it may control for banks with a different business model, or simply capture that securities holdings have different capital requirements, market-imposed borrowing constraints, or fair-value accounting requirements, which may affect bank capital structure (as discussed by Laux and Rauter 2017). Finally, $(Deposits/debt liabilities)_{i,t}$ controls for the fact that banks with greater access to wholesale (nondeposit) funding may have greater leverage procyclicality (as shown by Damar, Meh, and Terajima 2013).

Table 5, Columns 5 through 8, reports specifications with these additional variables interacted with *credit expansion*. The coefficients on $I[\text{Multiple branches}]_{i,t}$ and $(RWA/assets)_{i,t}$ are statistically significant and large in magnitude (i.e., 1-standard-deviation increase in *credit expansion* is associated

²⁰ Online Appendix Figure 5 reports estimates from an alternative specification in which Fitch support indicators are interacted with the bank size indicators, thus comparing banks of different Fitch support ratings *within* size groups. This specification helps to compare differences in perceived guarantee between otherwise similar banks. These results show that, even within size groups, the Fitch support indicator is associated with increased countercyclical equity issuance.

with a statistically significant 3.6- and 0.9-percentage-point decrease in *net equity issuance minus dividends*, conditional on the bank having 10+ branches and a 1-standard-deviation increase in risk-weighted assets to assets, respectively). This suggests that asset diversification across geographical regions (as in Demsetz and Strahan 1997) and increased risk-taking on the asset side are associated with greater equity payouts during credit expansions. However, Table 5 shows a small in magnitude and not statistically significant association for the other variables, suggesting that foreign branches, securities holdings, and access to nondeposit funding are not associated with equity issuance and payouts over the cycle, after controlling for the other variables in the regression.

Most importantly in Columns 5 through 8, the coefficient on $1[Fitch\ support \leq 3]_{i,t}$ remains similar in magnitude and statistically significant, even after the addition of all these other variables, suggesting that Fitch support ratings—representing expectations of government guarantees to creditors—capture a distinct channel affecting equity issuance, *alongside* these other channels. Also, from Table 5, the coefficients on the bank size indicators are negligible in magnitude and not statistically significant in Column 8, after including all the above additional variables, demonstrating that the size factor can mostly be explained by variables including government guarantees to creditors and these other factors.

4. Evidence from Landesbanken Subject to Removal of Creditor Guarantees

While the evidence in the previous section implicates the importance of government guarantees, one may be concerned about potential endogeneity. For example, there might be other important differences between banks with and without expectations of government support that account for differences in equity issuance.²¹ This section therefore studies a quasi-experimental setup, in which I exploit the surprise removal of government guarantees to German Landesbanken creditors.

I start by providing some brief institutional background on the Landesbanken and the court ruling that led to their loss of creditor guarantees. I then test a key mechanism of the model, that government guarantees dull creditor market discipline in response to equity issuance and payout announcements. The next part looks at the response of the German Landesbanken, which were differentially affected by the removal of creditor guarantees, and finds that the banks most affected by the European Commission ruling (i.e., those that lose a greater fraction of creditor guarantees) raise more equity and

²¹ For example, Hanson, Kashyap, and Stein (2011) suggest that the largest banks mainly compete on net interest margins and the cost of funding, whereas small banks focus on relationship lending, which can lead large banks to take on increased leverage to lower their average funding costs.

reduce payouts over the subsequent credit expansion. These results suggest that creditor guarantees have a meaningful impact on the decisions of bankers to issue new equity during credit expansions.

4.1 Institutional background

The following information is mainly taken from Gropp, Gruendl, and Guettler (2013) and Fischer et al. (2014) (the interested reader can refer to both for additional information).

Public savings banks in Germany are each associated with a Landesbank, and each of the Landesbanken is affiliated with a German federal state or group of states. The Landesbanken are internationally operating wholesale and investment banks that both lend to corporate customers (but not individuals) and act as liquidity clearinghouses for their member savings banks. Landesbanken common equity is held both by their member states and, to a considerable extent in the last two decades, by private investors, although shares are not publicly traded. Thus, they issue (and occasionally repurchase) common equity to private investors and pay out dividends. Unlike German savings and commercial banks, their main source of funding is long-term bonds, which until 2001 were explicitly guaranteed by their affiliated states, as discussed below.

Until the 2001 ruling, both the savings banks and Landesbanken had two types of government guarantees: an explicit guarantee of all of bank liabilities (“Gewährträgerhaftung”) and a maintenance obligation to keep banks sufficiently capitalized to survive and conduct business (“Anstaltslast”). The banks’ default risk was thus perceived to be low, which provided a competitive advantage through high credit ratings and low funding costs. German commercial banks, which felt themselves at a competitive disadvantage, tried repeatedly without success to remove these guarantees. Finally, the commercial banks in 1999 argued to the European Commission that the guarantees constituted state aid, violating Article 47 of the European Union treaty, which led to a formal investigation of their legality.

On July 17, 2001, bank representatives, the German Ministry of Finance, and the European Commission reached a compromise, which took investors by surprise. The ruling, known as the “Brussels Agreement,” was designed to phase out the guarantees. It stipulated that liabilities issued between July 19, 2001, and July 18, 2005, and maturing no later than December 31, 2015, were still covered by the guarantor’s liability, while those issued until July 18, 2005, but maturing after December 31, 2015, would not be guaranteed. Liabilities issued before the July 18, 2001, agreement were grandfathered in and maintained their guarantees, regardless of their maturities.

I exploit this setup in two ways. First, it allows me to look at the reaction of similar bonds of the same bank—some guaranteed (because their maturity is just before December 31, 2015), others not guaranteed (because their maturity is just after December 31, 2015), but having similar seniority, modified duration, and

other characteristics. Second, it allows me to study which banks lose guarantees at a faster rate (as a fraction of total debt). The reason is that the rate at which banks lose guarantees depends on idiosyncratic differences in the maturity structure of debt issued before the 2001 ruling: banks with longer maturity debt pre-2001 have their guarantees phased out more slowly because of the longer time before they have to rollover their debt to nonguaranteed debt. It is important to note that the 2001 European Commission ruling was a surprise, so that the maturity of debt issued pre-2001 could not have been influenced by foreknowledge of how the government guarantees were to be phased out.²²

For two reasons, I study the Landesbanken, but not the German public savings banks (the “Sparkassen”), which also lost their guarantees at the same time according to the same rules. First, the German savings banks do not have shareholders or raise equity, so it would be impossible to study equity issuance; nor do they issue bank debt (beyond issuing covered mortgage bonds), which would make it impossible to study creditor market discipline. Second, the savings banks were considerably less affected by the ruling because, unlike the Landesbanken, which are partly bond financed, they are mainly financed through deposits, which remained covered by deposit insurance.

4.2 Creditor market discipline following the removal of guarantees

Using the institutional setup described in the previous section, I now turn to testing a key element of the mechanism illustrated in the model in Section 1, that government guarantees dull creditor market discipline in response to equity issuance and payout announcements.

Recall from the model that, in the absence of creditor guarantees, creditors threaten to raise debt funding costs (which I refer to as “creditor market discipline”) in the event of equity payouts, given that equity payouts would increase the probability of firm bankruptcy. Conversely, they promise to lower debt funding costs in the event of new equity issuance, given that new equity helps minimize the probability of bankruptcy. However, in the presence of government guarantees to creditors, the model shows how creditor market discipline is blunted, allowing bankers to increase risk-shifting during asset expansions, by reducing equity issuance and increasing payouts, which would lead to the patterns of countercyclical equity issuance present in the data. The implication of these results is that, in the presence of strong government guarantees to creditors, new equity is perceived as “expensive” by bankers

²² Two caveats regarding the loss of creditor guarantees should be mentioned. First, the Landesbanken remained state owned and systemically relevant, because of their size and importance, so their debt may have still benefited from implicit guarantees. Second, as might be expected, the Landesbanken substantially increased their issuance of guaranteed debt just prior to the full phase out of guarantees in 2005, as a way to lock in lower funding costs. Although this implies that the Landesbanken had some discretion over the rate at which their guarantees are phased out, their ability to issue last-minute debt had limits (e.g., had to be balanced against the cost of holding excess liquidity).

during credit expansions, because issuance dilutes the option value of equity without lowering debt funding costs.²³

To test this key part of the mechanism, I start by showing that subordinated debt yields are sensitive to equity issuance and payout announcements both for U.S. commercial banks and for the German Landesbanken. I start with the U.S. case to establish broader relevance and external validity and then turn to the German Landesbank case to get sharp identification. Table 6 reports results for the U.S. case and shows the reaction of unsecured subordinated bank debt to common equity issuance and payout announcements. Comparing the change in yields from -3 days to $+3$ days around announcements, panel A, which looks at the twenty largest U.S. commercial banks from 1996 to 2015, shows that equity issuance announcements lead to lower yields (a statistically significant 11.5-bp decrease per equity issuance announcement), and vice versa for payout announcements (a statistically significant 1.6-bp increase per dividend increase announcement).

However, these magnitudes are not especially large in absolute size. This may be due to the fact that, as suggested by the mechanism, creditor market reactions might be blunted for the largest banks with presumably the largest implicit credit guarantees. To test this idea, panel B excludes the six largest banks (Citigroup, J. P. Morgan, Bank of America, Wells Fargo, State Street, and BNY Mellon), which may benefit the most from implicit guarantees to creditors. Consistent with the hypothesis, results are much stronger in this case. The reactions are larger in magnitude and statistically significant. In particular, equity issuance announcements are associated with a 22.9-bp drop in marginal funding costs per equity issuance announcement. (Online Appendix Figure 3 visualizes this event study, comparing yield changes subsequent to equity issuance for the six largest banks versus all other large banks, and verifies that the timing of the creditor reaction immediately coincides with the announcement.) As a robustness check, panel C additionally excludes the financial crisis period 2008–2010 because of high volatility in bond yields and because equity issuance during this time may have had unusually strong market reactions. Results in panel C, which excludes 2008–2010, are weaker in magnitude but qualitatively similar.

Next, I demonstrate the causal effect of the removal of government guarantees for the German Landesbanken, which leads creditor markets to be more sensitive to equity issuance and payouts after government guarantees are removed. Exogenous variation comes from comparing specific bond issues

²³ It is important to contrast the predictions of this mechanism with those of a standard signaling story, which makes the opposite predictions. In the signaling literature used to explain equity market reactions to equity issuance and payouts (e.g., Masulis and Korwar 1986), payouts are followed by positive stock price reactions (payouts being a positive signal about future firm earnings) and issuance by negative stock price reactions (issuance being a negative signal about future firm earnings). If equity issuance and payouts are a signal about future firm earnings, then bond price reactions also would be in the same direction (i.e., equity payouts would lower bond yields, while equity issuance would increase them). However, the mechanism hypothesized in this paper makes the opposite predictions for bonds, consistent with a shareholder-debtholder conflict rather than a signaling story: equity issuance (being a positive event for debt holders) is associated with a decrease in bond yield.

Table 6
Reaction of creditors to equity issuance: U.S. large commercial banks

A. Entire sample: U.S. large commercial banks

	Mean yield change (SE), $t = -3$ to $+3$
Issuance (N=18)	-0.115 (0.064)*
Dividend cut (N=33)	0.002 (0.079)
Dividend increase (N=215)	0.016 (0.006)***
Repurchase (N=51)	0.003 (0.01)
All (N=317) [after reversing sign for payouts]	-0.018 (0.01)*

B. Excluding six largest banks by assets

	Mean yield change (SE), $t = -3$ to $+3$
Issuance (N=14)	-0.229 (0.072)***
Dividend cut (N=19)	-0.140 (0.12)
Dividend increase (N=145)	0.026 (0.008)***
Repurchase (N=28)	0.029 (0.014)**
All (N=206) [after reversing sign for payouts]	-0.051 (0.014)***

C. Also excluding 2008-2010

	Mean yield change (SE), $t = -3$ to $+3$
Issuance (N=2)	-0.101 (0.107)
Dividend cut (N=2)	-0.121 (0.027)
Dividend increase (N=140)	0.024 (0.007)***
Repurchase (N=28)	0.029 (0.014)**
All (N=172) [after reversing sign for payouts]	-0.026 (0.006)***

This table reports the reaction of unsecured subordinated bank debt to common equity issuance and payout announcements. It shows that equity issuance announcements lead to lower yields, and vice versa for payout announcements. The table reports the average yield change (and standard error) in percentage points, before and after the announcement (± 3 days), across all common equity issuance and payout announcements of the twenty largest U.S. commercial banks over the period 1996-2015. Panel A is the full sample; panel B excludes the six largest U.S. banks, which may benefit the most from implicit guarantees to creditors; and panel C additionally excludes the financial crisis period 2008-2010. * $p < .1$; ** $p < .05$; *** $p < .01$.

of the same bank that were grandfathered into the old system and thus had government guarantees, versus newer bond issues (of similar seniority, modified duration, and other characteristics) that did not, and comparing the reaction of debt yields to equity issuance and payout announcements. This strategy makes use of a discontinuity across thresholds: for example, for each bank, I generally match unsecured subordinated debt issued between 2001 and 2005 that matures in late 2015 (and is thus *guaranteed* under the “Brussels Agreement”) with unsecured subordinated debt that matures in early 2016 (and is thus *not guaranteed*). Specifically, bonds of the same bank (one guaranteed, the other not guaranteed) are first 100% matched on seniority and other characteristics²⁴ and then on modified duration (based on their closest match). Online Appendix Section C.1 provides a full discussion of the matching procedure, including potential biases. Online Appendix Table 7 lists each announcement, a comparison of the bond issues, and their matched characteristics. Online Appendix Table 8 calculates the average difference in yields between matched bonds and shows that the average yield on guaranteed bonds is 1.5 to 3.5 percentage points lower than on nonguaranteed bonds.

²⁴ The other characteristics are fixed coupon, Euro-denominated, and not issued by a foreign subsidiary or special purpose entity of the bank.

Table 7
Reaction of creditors to equity issuance: German Landesbanken subject to removal of creditor guarantees

<i>Reaction of creditors to equity issuance and payouts (yield change in % points)</i>	
Difference: Nonguaranteed minus guaranteed	Mean (SE)
Issuance (N=13)	-0.284 (0.17)*
Issuance, excluding an outlier (N=12)	-0.103 (0.025)***
Dividend decrease (N=2)	-0.144 (0.156)
Dividend increase (N=11)	0.033 (0.02)*
Repurchase (N=2)	0.075 (0.025)
All (N=28) [after reversing sign for payouts]	-0.16 (0.084)**
All, excluding an outlier (N=27)	-0.075 (0.017)***

*** denote significance at the 10%, 5%, and 1% levels, respectively
This table reports the reaction of unsecured subordinated bank debt to common equity issuance and payout announcements, comparing government guaranteed debt versus nonguaranteed debt of the same bank with similar seniority, modified duration, and other characteristics. (The matching process is described in Online Appendix Section C.1, and Online Appendix Table 7 reports characteristics of the matched bonds.) In the absence of government guarantees to creditors, equity issuance announcements lead to decreased yields of unsecured subordinated bank debt, and vice versa for payout announcements. In contrast, guaranteed bank debt is insensitive to equity issuance and payouts. The table reports the average yield change (and standard error) in percentage points, before and after the announcement (± 3 days), across all common equity issuance and payout announcements from CapitalIQ of the eleven German Landesbanken subject to removal of creditor guarantees.
* $p < .1$; ** $p < .05$; *** $p < .01$.

Table 7 reports the yield reaction of unsecured subordinated bank debt to common equity issuance and payout announcements, comparing the reaction of guaranteed debt to nonguaranteed debt. As with the U.S. case in Table 6, in the absence of government guarantees to creditors, equity issuance announcements lead to decreased yields of unsecured subordinated bank debt, and vice versa for payout announcements. In contrast, guaranteed bank debt is insensitive to equity issuance and payouts. The magnitudes are meaningful, a difference of 28.4 bps (between nonguaranteed and guaranteed debt) in the drop in marginal funding costs subsequent to each equity issuance. Online Appendix Figure 4 visualizes the differential reaction between guaranteed versus nonguaranteed debt. The figure verifies the “parallel trends” assumption that guaranteed bank debt is insensitive to equity issuance and payouts, whereas nonguaranteed debt is immediately affected upon the announcement day. I conclude that, in the presence of credit guarantees, equity issuance is not compensated with a decrease in debt funding costs, which helps explain why equity issuance is perceived in this case as costly to bankers and their shareholders.

4.3 Landesbank equity issuance and payouts following the removal of creditor guarantees

Lastly, I look at the response of the German Landesbanken, analyzing equity issuance and payouts following their loss of creditor guarantees as a result of the July 17, 2001, ruling by the European Commission. I test the hypothesis that banks most affected by the ruling (i.e., those that lose a greater fraction of their creditor guarantees) raise more equity and reduce payouts over the subsequent Landesbank asset expansion (relative to those that are less affected

by the European Commission ruling).²⁵ Exogenous variation comes from the fact that bonds issued before the ruling were grandfathered in and bonds issued between July 2001 and July 2005 continued to have government guarantees depending on the rules specified in the 2001 “Brussels Agreement.” Therefore, the rate at which guaranteed debt is phased out for a given bank depends on idiosyncratic differences with the maturity structure of their debt just before the 2001 agreement: banks with longer maturity debt pre-2001 have their guarantees phased out more slowly because of the longer time before they have to rollover their debt. This fact introduces cross-sectional differences across Landesbanken in their fraction of government guaranteed debt over the subsequent credit cycle. As shown in Online Appendix Figure 5, some Landesbanken (e.g., LB Saar) lose over 20% of their debt guarantees by the end of 2005 (and 35% by the end of 2006), whereas other Landesbanken lose almost no guarantees (e.g., Bremer LB).

One assumption of this identification strategy is that, after the 2001 ruling, the Landesbanken and their creditors could predict the effect it would have on debt guarantees for individual banks in subsequent years. However, this seems a realistic assumption given (1) computing this information based on the maturity structure of each bank’s debt (as done in this paper and shown in Online Appendix Figure 5) is straightforward; and (2) the loss of government guarantees and their consequences were widely analyzed at the time by the ratings agencies (see, e.g., the discussion in Fischer et al. 2014). In fact, ratings agencies (who arguably work on behalf of creditors and are, in practice, a main enforcer of “creditor market discipline”) often threatened to downgrade Landesbank debt subsequent to the guarantee removal, urging the Landesbanken to improve their profitability and strengthen their balance sheets through recapitalization.

To analyze the effect of the removal of government guarantees on subsequent equity issuance and payouts, I estimate the following cross-sectional regression:

$$y_i = \alpha + \delta(\text{decrease in guaranteed debt})_i + \varepsilon_i. \quad (2)$$

The independent variable is the decrease in the fraction of guaranteed debt outstanding for each bank from the end of 2000 (just before the ruling) to the end of 2005 (after the creditor guarantees were fully phased out for newly issued debt). It is important to note that the results in Table 8 are robust to measuring the fraction of guaranteed debt in other years after 2005, because the phase out of creditor guarantees over time is persistent and generally rank preserving among banks, as shown in Online Appendix Figure 5.

²⁵ One potential complication for this analysis is the possibility that the Landesbank credit expansion was driven by the rush to issue new guaranteed debt and to turn it into credit and other investments before the July 2005 deadline. However, Online Appendix Figure 7 presents evidence suggesting that this force was unlikely to be the main driver of the Landesbank credit expansion, a conclusion echoed by Fischer et al. (2014). See the discussion in Online Appendix Section C.2.

Table 8
Landesbank equity issuance following removal of creditor guarantees

A. Landesbank equity issuance and payouts (2001-2005)									
	Net equity issuance minus dividends			Equity issuance	Dividends	Net income minus payouts			
	(1)			(2)	(3)				(4)
Δ (Decrease in fraction of guaranteed debt) _i	2.427**			2.336*	-0.092			1.357**	
p-value (t-test, Huber-White SE)	[.025]			[.028]	[.164]			[.005]	
p-value (wild bootstrap)	[.025]			[.061]	[.279]			[.040]	
Adj. R ²	.53			.49	.03			.63	
N	11			11	11			11	
B. Landesbank risk-taking (2001-2005) and post-crisis equity increase (2008-2012)									
	Risk-adjusted		Change in (Liquid	Nonperforming	Equity issuance,	Equity issuance plus additions to			
	assets to assets		assets / assets)	loans, peak	2008 to 2012			ret. earn., 2008 to 2012	
	(1)		(2)	(3)	(4)			(5)	
Δ (Decrease in fraction of guaranteed debt) _i	-2.789**		1.526*	-0.464	-2.481			-4.459*	
p-value (t-test, Huber-White SE)	[.003]		[.016]	[.182]	[.296]			[.038]	
p-value (wild bootstrap)	[.023]		[.053]	[.250]	[.309]			[.098]	
Adj. R ²	.54		.32	.05	-.04			.43	
N	10		11	9	9			9	

This table analyzes equity issuance, payouts, and risk-taking of eleven German Landesbanken following their loss of creditor guarantees as a result of the July 17, 2001, ruling by the European Commission. Specifically, the table reports cross-sectional regression estimates from Equation (2). The independent variable is the decrease in the fraction of guaranteed debt outstanding for each bank between the end of 2000 (just before the ruling) and the end of 2005 (after the creditor guarantees were fully phased out for newly issued debt). In panel A, the dependent variables are net issuance minus dividends (Column 1), new equity issuance (Column 2), dividends (Column 3), and changes in retained earnings (net income minus payouts, Column 4); these four variables are computed as the cumulative change (as fractions of book equity) from the end of 2001 (just after the ruling) to the end of 2005, a period which roughly corresponds to a Landesbank asset expansion. In panel B, the dependent variables are risk-adjusted assets to assets (Column 1), the percentage-point change in liquid assets to assets from 2001 to 2005 (Column 2), peak nonperforming loans over 2008 to 2012 (Column 3), *post-crisis* new equity issuance (Column 4), and new equity issuance plus additions to retained earnings (Column 5) over 2008-2012 (note: two Landesbanken exited the sample in 2008 due to forced mergers). Given the small number of observations, *p*-values are estimated from wild bootstrapped standard errors. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 8 reports cross-sectional regression estimates from Equation (2). In panel A, the dependent variable in Equation (2) can be variously *net equity issuance*, *dividends*, *net equity issuance minus dividends*, or *net income minus payouts* (i.e., changes in retained earnings). To capture how banks raise and pay out equity during credit expansions, these four dependent variables (normalized by 2002 book equity) are computed as cumulative changes from 2002 (just after the ruling) to the end of 2005 (after the creditor guarantees were fully phased out for newly issued debt), a period which also roughly corresponds to a Landesbank credit expansion. I take the end of 2005, rather than 2006 or 2007, as the end cutoff, so that it coincides with the time period of the independent variable, the fraction of guaranteed debt. It is important to point out that the period 2000 to 2005 corresponds to a modest credit expansion of the German Landesbanken sector and of the Eurozone banking sector as a whole, even though there was not an aggregate credit expansion in Germany over this period.²⁶

Turning first to panel A, one sees that the loss of creditor guarantees leads banks to raise and retain substantially more equity over the subsequent credit expansion. Although the sample size is small, corresponding to just eleven German Landesbanken, the results are strong and statistically significant for all the dependent variables except for dividends. Because of the small sample size, I estimate Wild bootstrap standard errors, which MacKinnon (2013) finds perform best in terms of power for small samples.

To give a sense of the magnitudes, Table 8, panel A, shows that when the fraction of guaranteed debt decreases, for example, by 10 percentage points, cumulative equity issuance from 2002 to 2005 increases by 23.36 percentage points of book equity, and retained earnings increase by 13.57 percentage points of book equity. Figure 4 visualizes the results. The figure shows the strong relationship between the fraction of guaranteed debt and equity issuance. These large recapitalizations, both through new issuance and increased equity retention, thus appear to be spurred by creditors' (and ratings agencies') demands in response to the loss of government guarantees. These results suggest

²⁶ Online Appendix Figure 6 documents that Landesbank loans increased on average by 25% by 2005 (relative to 2000 levels) and by almost 50% by 2008, with similar increases in average Landesbank total assets. The reason for the credit expansion among the Landesbanken, but not German commercial banks, during this period can be explained by both the geographical exposure and the business models of the Landesbanken. According to Online Appendix Table 9, only 54% of Landesbanks' asset exposure was in Germany, with the large majority of their remaining exposure spread across the rest of Europe. Additionally, according to Online Appendix Table 9, around ~33% of their assets consisted of securities—in particular, the Landesbanken were important buyers of U.S. and U.K. mortgage-backed securities and asset-based commercial paper over this period. Thus, the Landesbanken had exposure to the Eurozone-wide credit expansion outside of Germany, which explains why their balance sheets were growing during this period. Lastly, the Landesbanken were primarily corporate lenders, along with focusing on niche areas, such as shipping finance, areas that saw modest credit expansion, even in Germany. In contrast, the rest of the German banking sector is dominated by residential mortgage lending, which was contracting during this period.

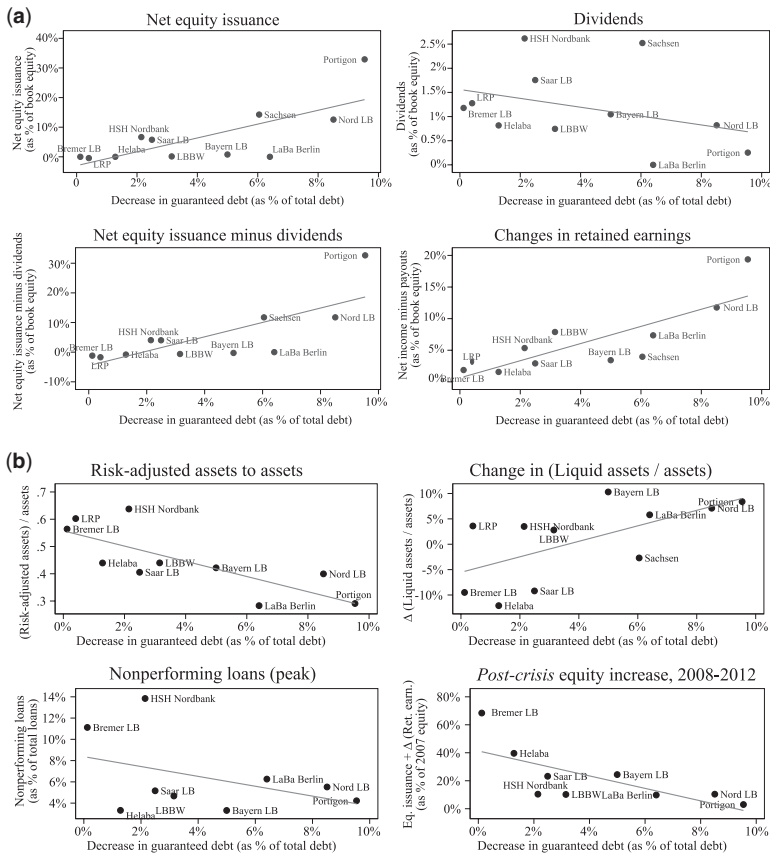


Figure 4
Landeskbank equity issuance and payouts following removal of creditor guarantees

This figure plots various measures of equity issuance and payouts of eleven German Landesbanken following their loss of creditor guarantees as a result of the July 17, 2001, ruling by the European Commission. The variable plotted on the x-axis is the percentage-point decrease in guaranteed debt outstanding for each bank from the end of 2000 (just before the ruling) to the end of 2005 (after the creditor guarantees were fully phased out for newly issued debt). In panel A, the variables plotted on the y-axis are net issuance of common equity (top left), dividends (top right), net issuance of common equity minus dividends (bottom left), and changes in retained earnings (net income minus payouts to common equity, bottom right); these four variables are computed as cumulative changes (as a percentage of 2001 book equity) from the end of 2001 (just after the ruling) to the end of 2005, a period that roughly corresponds to a Landesbank asset expansion (see Online Appendix Figure 6). In panel B, the variables plotted on the y-axis are: risk-adjusted assets to assets (top left), the percentage-point change in liquid assets to assets from 2001 to 2005 (top right), peak nonperforming loans over 2008 to 2012 (bottom left), and the *post-crisis* equity increase over the period 2008-2012 (bottom right); note that for these last three variables, two Landesbanken exited the sample in 2008 because of forced mergers.

that the loss of creditor guarantees leads banks to raise and retain substantially more equity over the subsequent credit expansion.²⁷

²⁷ One potential concern is that these results may simply reflect “pecking-order theory”: when debt funding costs increase (because of the loss of government guarantees), banks will shift their capital structure toward more

5. Alternative Explanations for Countercyclical Equity Issuance

It is important to emphasize that bank motivations regarding equity issuance and payouts are likely multifactorial. While this paper focuses on government guarantees because of the magnitudes and policy implications, it is also important to examine alternative explanations regarding patterns of equity.

I start by testing two alternative explanations. The first is that poor shareholder governance may lead banks to take excessive leverage (as suggested by Morellec, Nikolov, and Schurhoff 2012) during credit expansions.²⁸ In the second, bank managerial behavioral biases, including overoptimism and short-termism during credit expansions, may lead bankers to avoid raising equity and increasing payouts, because they underestimate the risk of a future banking crisis. Although multiple managerial behavioral biases might be relevant, I focus on one that I can test, bank CEO overoptimism, using data from Ma (2015) on CEOs' holding decisions of their own banks' equity. I lastly discuss evidence regarding other potential explanations for countercyclical equity issuance.

5.1 Bank shareholder governance

Table 9 reports results estimated from Equation (1), with *net equity issuance minus dividends* as the dependent variable. As in Table 2, *credit expansion* is the main independent variable, which is interacted with bank size indicator variables in various specifications. Two additional variables proxy for the quality of shareholder governance: the GIM corporate governance index (Gompers, Ishii, and Metrick 2003) and the E governance index (Bebchuk, Cohen, and Ferrell 2009).²⁹

Columns 1–4 report results for the GIM governance index, and Columns 5–8 for the E governance index. Both governance indexes show that weaker shareholder governance (a higher index value) is associated with increased

equity simply because of lower costs of funding. I therefore test the following stronger additional predictions that emerge from a model of agency conflicts between shareholders and creditors: (1) banks that lose their guarantees more quickly also decrease risk on the asset side, and (2) banks that raise more equity during the credit expansion due to their loss of guarantees will raise *less* after the bust (i.e., have a more procyclical pattern). Table 8, panel B, tests these additional predictions from the model. Columns 1 through 3 show that banks that lose their guarantees more quickly take on less asset risk *ex ante* (as measured by decreased risk-adjusted assets to assets and an increased fraction of liquid assets) and *ex post* (as measured by peak nonperforming loans over the period 2008 to 2012). Columns 4 and 5 show that banks that lose their creditor guarantees more quickly need to raise less equity after the bust (in 2008 to 2012), establishing a more procyclical equity pattern for these banks. These findings confirm some of the predictions of the model and differentiate the mechanism from a pure pecking order story.

²⁸ Alternatively, the effect could go in the other direction, as banks with more shareholder-friendly boards and better-aligned incentives between management and shareholders may actually take more risk, as shown by Fahlenbrach and Stulz (2011) and Beltratti and Stulz (2012), because exploiting government guarantees is good for shareholders.

²⁹ The GIM index combines twenty-four unique governance rules to construct a proxy for the level of shareholder rights over the period 1990–2006. A higher value corresponds to weaker shareholder rights. The “entrenchment” (or “E”) index is a subset of six measures from the GIM governance index, which Bebchuk et al. (2009) argue are the most economically significant.

Table 9
Governance and CEO overoptimism

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Δ (Bank credit / GDP) _{<i>it</i>}	-0.025*** [-4.499]	-0.025*** [-4.533]			-0.026*** [-4.541]	-0.026*** [-4.590]			-0.024*** [-3.652]	-0.016*** [-2.304]		
Δ (Bank credit / GDP) _{<i>it</i>} × GIM governance index		-0.006 [-1.623]		-0.006** [-2.071]								
× E governance index						-0.004 [-1.396]		-0.008*** [-2.724]				
× CEO optimism										-0.012* [-1.668]		-0.011 [-1.471]
× 1[#1-5 by assets] _{<i>it</i>}			-0.043*** [-2.703]	-0.047*** [-3.061]			-0.050*** [-3.045]	-0.062*** [-3.594]			-0.054 [-1.603]	-0.046 [-1.357]
× 1[#6-20 by assets] _{<i>it</i>}			-0.026*** [-3.189]	-0.025*** [-3.120]			-0.027*** [-3.276]	-0.031*** [-3.582]			-0.044*** [-3.216]	-0.035*** [-2.289]
× 1[#21-100 by assets] _{<i>it</i>}			-0.024*** [-3.479]	-0.023*** [-3.406]			-0.024*** [-3.457]	-0.023*** [-3.312]			-0.020* [-1.876]	-0.013 [-1.141]
× 1[#101-500 by assets] _{<i>it</i>}			-0.008*** [-0.898]	-0.015*** [-1.170]			-0.008*** [-0.967]	-0.008 [-1.027]			-0.024*** [-3.225]	-0.016*** [-2.349]
× 1[#501+ by assets] _{<i>it</i>}			-0.008*** [-2.893]	-0.015*** [-3.763]			-0.009*** [-3.031]	-0.020*** [-4.328]			0.002 [0.107]	0.011 [0.539]
Constant	0.006 [0.689]	0.006 [0.695]	0.061 [0.991]	0.054 [0.887]	0.005 [0.599]	0.006 [0.613]	0.056 [0.829]	0.051 [0.769]	0.009 [1.493]	0.008 [1.476]	0.016 [0.858]	0.015 [0.816]

Bank-quarter controls Yes Yes Yes Yes Yes Yes Yes Yes Yes Yes Yes Yes Yes
Bank size indicator controls Yes Yes Yes Yes Yes Yes Yes Yes Yes Yes Yes Yes Yes
Bank fixed effects .04 .04 .05 .05 .04 .04 .04 .04 .04 .05 .05 .05 .05
Adj. R² 8.668 8.668 8.668 8.668 8.350 8.350 8.350 8.350 8.350 8.602 8.602 8.602 8.602
N This table is similar to Table 2, panel A, but includes the following additional variables: $GIM\ governance\ index_{i,t}$, the corporate governance index from Gompers, Ishii, and Metrick (2003); $E\ governance\ index_{i,t}$, the corporate governance index from Bebchuk, Cohen, and Ferrell (2009); and $CEO\ optimism_{i,t}$, an indicator variable of bank CEO overoptimism from Ma (2015) based on a bank CEO's exercise of personal stock options. As in Table 2, panel A, the dependent variable is net issuance of common equity minus dividends, and bank-level control variables (coefficient estimates not reported) are $(Income/book\ equity)_{i,t}$, the previous year's net income (as a fraction of the previous year's book equity); $(Market/book)_{i,t}$, the beginning-of-quarter market-to-book ratio; $(Previous\ year's\ stock\ return)_{i,t}$, the previous year's total return of the bank's stock price; and $1[Undercapitalized]_{i,t}$, an indicator variable that equals 1 when the previous year's equity-to-assets ratio of a bank is less than 5%. Bank size indicator variables (e.g., $1[\#1-5\ by\ assets]_{i,t}$) outside the interaction term are included (but not reported for space reasons) in the specifications in Columns 3-4, 7-8, and 11-12. All continuous independent variables are expressed in units of standard deviations. Observations are quarterly. The *t*-statistics are computed from standard errors double-clustered on firm and quarter. * $p < .1$; ** $p < .05$; *** $p < .01$.

countercyclical equity issuance, though the coefficient is quite small in magnitude. In addition, the governance indexes do not diminish the magnitude of the coefficients on the bank size interaction variables. In fact, they increase it. I thus conclude that governance indexes do not seem to explain the size factor related to the cyclicity of equity issuance.

5.2 Bank CEO overoptimism

If bankers are overoptimistic during credit expansions, more so than stock market investors, this might lead them to repurchase their firms' equity, believing it to be underpriced. This would lead to countercyclical equity issuance. To test this hypothesis, I use the options-based measure of bank CEO optimism provided by Ma (2015), which is based on whether CEOs choose to exercise vested deeply-in-the-money options, following Malmendier and Tate (2005). The theoretical framework is provided by Hall and Murphy (2002): because CEOs generally have large equity positions in their firms, underdiversification and risk aversion make it optimal for a rational CEO to exercise options that are enough in-the-money immediately after vesting.³⁰

Columns 9–12 in Table 9 show the estimation results with bank size interaction variables, along with the “CEO optimism” indicator interacted with *credit expansion*. I find, consistent with the hypothesis, that higher CEO optimism is associated with stronger countercyclical equity issuance. However, like with the governance indexes, the “CEO optimism” indicator does not diminish the magnitude of coefficients on the bank size interaction variables. Thus, CEO optimism seems to be a separate channel, which does not explain the size factor related to the cyclicity of bank equity issuance.

5.3 Other potential explanations for countercyclical equity issuance

In Online Appendix Section H, I bring together evidence—both from Section 3 and new evidence on patterns of bank equity issuance across countries and historical time periods—to evaluate other potential alternative explanations for countercyclical bank equity issuance. These other explanations include value-at-risk or return-on-equity targeting (Adrian and Shin 2013), market timing (Baker and Wurgler 2002), and other behavioral explanations. I argue that market timing has a difficult time explaining countercyclical equity issuance. For the other explanations, in general they have difficulty in explaining the variation in patterns of equity issuance (procyclical versus countercyclical) across banks of various sizes, across developed countries, and across historical time periods in the United States.

³⁰ The option variable, which I label in Table 9 as “CEO optimism,” is annual, and, following Ma (2015), equals 1 in a certain year if the CEO does not exercise any of his or her vested options that are more than 67% in-the-money in that year; it equals 0 if the CEO exercises at least some vested options that are more than 67% in-the-money; and it is set to missing if the CEO does not have any vested options that are more than 67% in-the-money in that year.

6. Conclusions

This paper finds that bank equity issuance and retained earnings are countercyclical across credit cycles over the period 1980–2012. Thus, during credit expansions, even though more capital might help banks better absorb shocks, banks raise and retain equity the least. A mechanism is proposed showing why government guarantees may lead banks to choose countercyclical equity issuance. Consistent with the model, evidence using Fitch support ratings suggests that bank equity issuance is countercyclical when government guarantees are strong and procyclical when government guarantees are weak or absent. The analysis then turns to the surprise removal of government guarantees to the German Landesbanken. Testing a key part of the mechanism, I find that government guarantees dull creditor market discipline in response to equity issuance and payout announcements. Lastly, I find that the banks most affected by the European Commission ruling raise more equity and reduce payouts over the subsequent credit expansion.

These results, taken together, suggest that large banks may resist raising equity because, with government guarantees, equity issuance is not compensated with a decrease in debt funding costs (and vice versa for payouts), making equity issuance costly for shareholders. As a result, government guarantees distort the incentives of banks to raise new equity and affect the dynamics of bank capital structure over the credit cycle. As a policy consequence, regulators may want to even further strengthen countercyclical capital buffers, especially for the largest banks, and design bail-in mechanisms and orderly resolution mechanisms that promote strong creditor market discipline *ex ante* (by credibly threatening junior creditors with haircuts in the event of bank default), while preventing catastrophic defaults that could lead to contagion.

References

- Acharya, V., D. Anginer, and J. Warburton. 2016. The end of market discipline? Investor expectations of implicit government guarantees. Working Paper, New York University.
- Acharya, V., I. Gujral, N. Kulkarni, and H. S. Shin. 2012. Dividends and bank capital in the financial crisis of 2007–2009. Working Paper, New York University.
- Acharya, V., H. Le, and H. S. Shin. 2017. Bank capital and dividend externalities. *Review of Financial Studies* 30:988–1018.
- Admati, A., P. DeMarzo, M. Hellwig, and P. Pfleiderer. 2013. Fallacies, irrelevant facts, and myths in the discussion of capital regulation: Why bank equity is not socially expensive. Working Paper, Stanford Graduate School of Business.
- Adrian, T., and H. S. Shin. 2013. Procyclical leverage and value-at-risk. *Review of Financial Studies* 27:373–403.
- Allen, F., E. Carletti, and R. Marquez. 2011. Credit market competition and capital regulation. *Review of Financial Studies* 24:983–1018.
- Baker, M., and J. Wurgler. 2002. Market timing and capital structure. *Journal of Finance* 57:1–32.

- Baron, M., and W. Xiong. 2017. Credit expansion and neglected crash risk. *Quarterly Journal of Economics* 132:713–64.
- Bebchuk, L., A. Cohen, and A. Ferrell. 2008. What matters in corporate governance? *Review of Financial Studies* 22:783–827.
- Beltratti, A., and R. Stulz. 2012. The credit crisis around the globe: Why did some banks perform better? *Journal of Financial Economics* 105:1–17.
- Calomiris, C., and C. Kahn. 1991. The role of demandable debt in structuring optimal banking arrangements. *American Economic Review* 81:497–513.
- Cheng, I., S. Raina, and W. Xiong. 2014. Wall Street and the housing bubble. *American Economic Review* 104:2797–829.
- Damar, E., C. Meh, and Y. Terajima. 2013. Leverage, balance-sheet size and wholesale funding. *Journal of Financial Intermediation* 22:639–62.
- DeAngelo, H., and R. Stulz. 2015. Liquid-claim production, risk management, and bank capital structure: Why high leverage is optimal for banks. *Journal of Financial Economics* 116:219–36.
- Demsetz, R., and P. Strahan. 1997. Diversification, size, and risk at bank holding companies. *Journal of Money, Credit and Banking* 29:300–313.
- Diamond, D., and R. Rajan. 2000. A theory of bank capital. *Journal of Finance* 55:2431–65.
- Diamond, D., and R. Rajan. 2001. Banks and liquidity. *American Economic Review* 91:422–5.
- Fahlenbrach, R., R. Prilmeier, and R. Stulz. 2017. Why does fast loan growth predict poor performance for banks? *Review of Financial Studies* 31:1014–63.
- Fahlenbrach, R., and R. Stulz. 2011. Bank CEO incentives and the credit crisis. *Journal of Financial Economics*. 99:11–26.
- Fischer, M., C. Hainz, J. Rocholl, and S. Steffen. 2014. Government guarantees and bank risk taking incentives. Working Paper, Goethe University Frankfurt.
- Gompers, P., J. Ishii, and A. Metrick. 2003. Corporate governance and equity prices. *Quarterly Journal of Economics* 118:107–56.
- Gropp, R., C. Gruendl, and A. Guettler. 2013. The impact of public guarantees on bank risk-taking: Evidence from a natural experiment *Review of Finance* 18:457–88.
- Gropp, R., H. Hakenes, and I. Schnabel. 2010. Competition, risk-shifting, and public bail-out policies. *Review of Financial Studies* 24:2084–120.
- Hall, B., and K. Murphy. 2002. Stock options for undiversified executives. *Journal of Accounting and Economics* 33:3–42.
- Hanson, S., A. Kashyap, and J. Stein. 2011. A macroprudential approach to financial regulation. *Journal of Economic Perspectives* 25:3–28.
- Holmstrom, B., and J. Tirole. 1997. Financial intermediation, loanable funds, and the real sector. *Quarterly Journal of Economics* 112:663–91.
- Jagannathan, M., C. Stephens, and M. Weisbach. 2000. Financial flexibility and the choice between dividends and stock repurchases. *Journal of Financial Economics* 57:355–84.
- Kashyap, A., and J. Stein. 2004. Cyclical implications of the Basel II capital standards. *Economic Perspectives-Federal Reserve Bank of Chicago* 28:18–33.
- Kashyap, A., R. Rajan, and J. Stein. 2002. Banks as liquidity providers: An explanation for the coexistence of lending and deposit taking. *Journal of Finance* 57:33–73.
- Kashyap, A., R. Rajan, and J. Stein. 2008. Rethinking capital regulation. In *Maintaining stability in a changing financial system*, 431–71. Kansas City, MO: Federal Reserve Bank of Kansas City.

- Laux, C., and T. Rauter. 2017. Procyclicality of U.S. bank leverage. *Journal of Accounting Research* 55:237–73.
- Ma, Y. 2015. Bank CEO optimism and the financial crisis. Working Paper, University of Chicago.
- MacKinnon, J. 2013. Thirty years of heteroskedasticity-robust inference. In *Recent advances and future directions in causality, prediction, and specification analysis*, eds. X. Chen and N. R. Swanson, 437–61. New York: Springer.
- Masulis, U., and G. Tate. 2005. CEO overconfidence and corporate investment. *Journal of Finance* 60:2661–700.
- Masulis, R., and A. Korwar. 1986. Seasoned equity offerings: An empirical investigation. *Journal of Financial Economics* 15:91–118.
- Mehran, H., and A. Thakor. 2010. Bank capital and value in the cross-section. *Review of Financial Studies* 24:1019–67.
- Mian, A., A. Sufi, and E. Verner. 2017. Household debt and business cycles worldwide. *Quarterly Journal of Economics* 132:1755–817.
- Miller, M. 1995. Do the M & M propositions apply to banks? *Journal of Banking and Finance* 19:483–89.
- Morellec, E., B. Nikolov, and N. Schürhoff. 2012. Corporate governance and capital structure dynamics. *Journal of Finance* 67:803–48.
- Poghosyan, T., C. Werger, and J. de Haan. 2016. Size and support ratings of US banks. *North American Journal of Economics and Finance* 37:236–47.
- Schularick, M., and A. Taylor. 2012. Credit booms gone bust: Monetary policy, leverage cycles, and financial crises, 1870–2008. *American Economic Review* 102:1029–61.
- Stephens, C., and M. Weisbach. 1998. Actual share reacquisitions in open market repurchase programs. *Journal of Finance* 53:313–33.